

**Preliminary Basis of Design Report  
Water  
For  
Finish Line Auto Club – Perimeter Center  
Princess Drive and Loop 101  
Scottsdale, Arizona 85255**

**COS Plan Check #: 1059-PA-2024  
COS Case #: 19-ZN-2004#2**

**1<sup>st</sup> Submittal 6/17/25**

**2<sup>nd</sup> Submittal 8/11/25**

**3<sup>rd</sup> Submittal 10/1/25**



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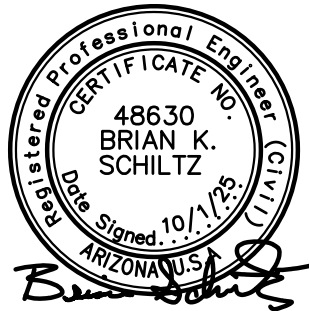
**H.E. PROJECT NO. LGEC332**

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## **1.0 Introduction**

This design report has been prepared under a contract from LGE Corporation, developer of the Finish Line Auto Club project. The purpose of this report is to provide a water analysis, required by the City of Scottsdale, to support this development. Preparation of this report has been done according to the procedures detailed in Chapter 6 of the City of Scottsdale's *Design Standards and Policy Manual* (Reference 1).

This development project is located along the west side of N. Perimeter Drive and along the east side of N. Pacesetter Way, between E. Pacesetter Way and E. Princess Drive, within the City of Scottsdale, Maricopa County, Arizona. The site is specifically located in the south half of Section 36, Township 4 North, Range 4 East of the Gila and Salt River Base and Meridian. Figure 1, in Appendix A, illustrates the location of the project site in relation to the City of Scottsdale street system. Access to the site is provided from N. Perimeter Drive and N. Pacesetter Way.

The proposed project site is currently a developed parcel within the Perimeter Center master development. The existing parcel is bound by existing commercial developments to the north and south, N. Perimeter Drive to the east and N. Pacesetter Way to the west.

The development proposes the construction of four new car condominium units, denoted as Buildings B through E, and a car condominium Clubhouse, denoted as Building A, on the easterly side of the project. This campus is dedicated to a private auto club. A car storage building is proposed on the westerly side of the parcel. This building totals 49,266 square feet and will be classified as a flex-industrial building. Site improvements will include construction of driveway entrances, parking lots, sidewalk/hardscape, landscape areas, and supporting infrastructure including new storm water drainage systems, water, sewer and fire line services. The overall project site is approximately 10.394± acres.

## **2.0 Existing Condition**

The existing site consists of a single commercial building and associated parking lots with parking canopies. The site is located in Pressure Zone 4. Adjacent to the site, there is an existing 8-inch DIP public water main along the east side of N. Pacesetter Way and an existing 12-inch DIP public water main along the east side of N. Perimeter Drive. There is an existing 8-inch public water line, enveloped by a 12-foot existing public water line easement, that runs through the site with a connection to the aforementioned existing public main in N. Perimeter Drive and extends to an existing 8-inch public main in E. Princess Drive. This existing 8-inch public water line currently services the existing building on the site and the existing building south of the site. There are also two existing fire hydrants connected to this existing 8-inch public water line.

### 3.0 Proposed Water System

Two new public 8-inch water lines will be looped through the site within 20-foot public water line easements and will provide fire protection along with domestic and landscape services. One of the water line loops will service the proposed car storage building and the other will service the four car condominium units and Clubhouse. The car storage building water line loop will have two connections to the existing 8-inch DIP public water main along the east side of N. Pacesetter Way. The water line loop servicing the four car condominium units and Clubhouse will have two connections to the existing 8-inch public water line that runs through the site within a 12-foot public water line easement. A portion of the existing onsite public water line and easement, along with one existing on-site fire hydrant, will be abandoned and removed to accommodate the proposed site plan. Respective domestic, landscape, and fire services will be taken from the new public waterline loops. Additionally, there are ten proposed onsite fire hydrants. The Conceptual Utility Plan, in Appendix A, depicts the proposed water improvements.

### 4.0 Projected System Demands

The Average Day Demand, Peak Hour Demand, and Maximum Day Demand for this development were derived using unit flow requirements from the City of Scottsdale *Design Standards & Policies Manual* for Water, Figure 6.1-2, see Appendix C. The proposed car storage building will be classified as a flex-industrial building. Therefore, per Figure 6.1-2, an average day water demand of 1.44 gallons per minute (GPM) per acre was used for the car storage building. For the four car condominium units and the Clubhouse, per Figure 6.1-2, an average day water demand of 0.00111 GPM per square foot was used. The site is currently a single parcel that will be divided between the proposed car storage building and the five car condominium units/Clubhouse. The proposed parcel line will be used to determine the site acreage for the car storage building domestic demands. As part of the car storage building construction, this building is expected to install 15 water closets and 13 sinks for a total of 86 fixture units per Table 709.1 of the 2018 International Plumbing Code. The complete water demand calculations for the proposed buildings can be found in Appendix B. The Maximum Day Demand is 2.0 times the Average Day Demand, and the Peak Hour Demand is 3.5 times the Average Day Demand. The total Average Day Demand for the development exceeds 100,000 gallons per day (GPD). Therefore, per City of Scottsdale requirements, a Water Demand Exhibit is shown in Appendix E.

The proposed car storage building is 49,266 square feet and will be type V-B construction. Between the four car condominium units and Clubhouse, the Clubhouse is the largest proposed building. The Clubhouse is 63,020 square feet and will also be type V-B construction. Per the 2021 International Fire Code, the minimum base fire flow rate for buildings of this size and construction type is 6,000 gallons per minute (GPM) and 6,750 GPM, respectively. Because the buildings will be protected by an approved sprinkler system, per NFPA 13, the required fire flow may be 25% of the table value, but not below 1,500 GPM. Fire flows of 1,500 GPM for the car storage building and 1,688

GPM for the Clubhouse are used for the site analysis. Per City of Scottsdale, pressure requirements, minimum acceptable design pressures are 30 psi at the hydrant under design fire flow requirements and 15 psi at all domestic demand nodes, and minimum residual pressure, while applying initial service line design flow, of 50 psi at highest finished floor for domestic demand.

## **5.0 Water Model**

A fire flow test was completed on April 14<sup>th</sup>, 2025 by Arizona Flow Testing, LLC and the results are shown in Appendix D. This test data was used to model the proposed system using WaterCad, a pipe network analysis program by Haestad Methods. A reservoir and pump were added to the model near the flow test location to simulate pressure versus flow curve. The hydrant test was adjusted per the City of Scottsdale requirements. Note that the pipes PX-1 and PX-2 connecting the pump and reservoir are not a part of the system and are oversized to 120-inches to minimize system losses. Pipes and junctions were added to the network model matching the pipe sizes, materials, and elevations of the proposed and existing system.

The fire flow model was set up such that 1,500 GPM and 1,688 GPM, for the proposed car storage building and Clubhouse, respectively, is taken from the hydrant furthest from the proposed connections to the existing public system. The model was analyzed using separate fire flows with the maximum day demands to identify the worst-case hydraulic node. The car storage building fire flow created the worst-case scenario. The lowest resultant pressure based upon the required fire flow and the maximum day demands was calculated to be 59.29 psi at node J-21. This is above the 30-psi fire flow minimum pressure. The lowest resultant pressure for the peak flow is 66.98 psi at node J-6, which is greater than the minimum peak flow pressure of 50 psi. Results and data from the WaterCAD model are shown in Appendix B.

## **6.0 Conclusion**

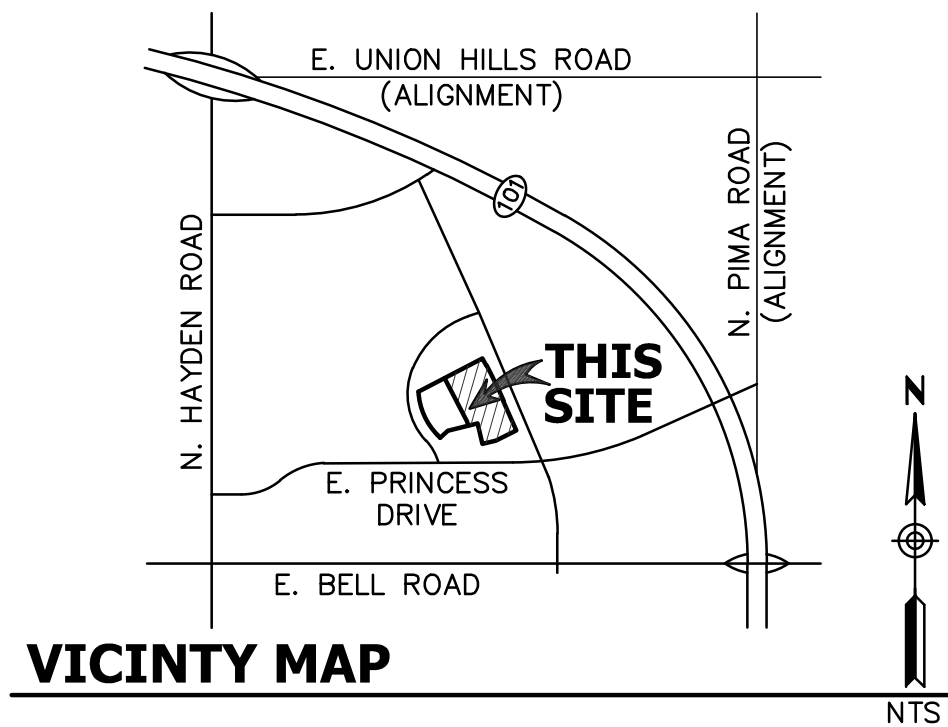
Based on the results of this study, it can be concluded that the existing and proposed water systems are adequate to support this development.

## **7.0 References**

- 1) City of Scottsdale Design Standard & Policies Manual, January 2018.
- 2) International Code Council International Fire Code (2021).

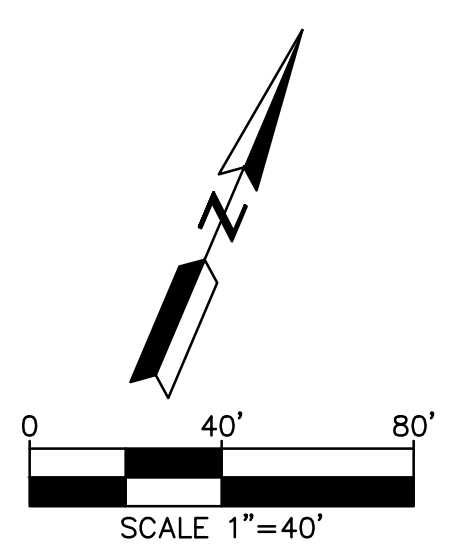
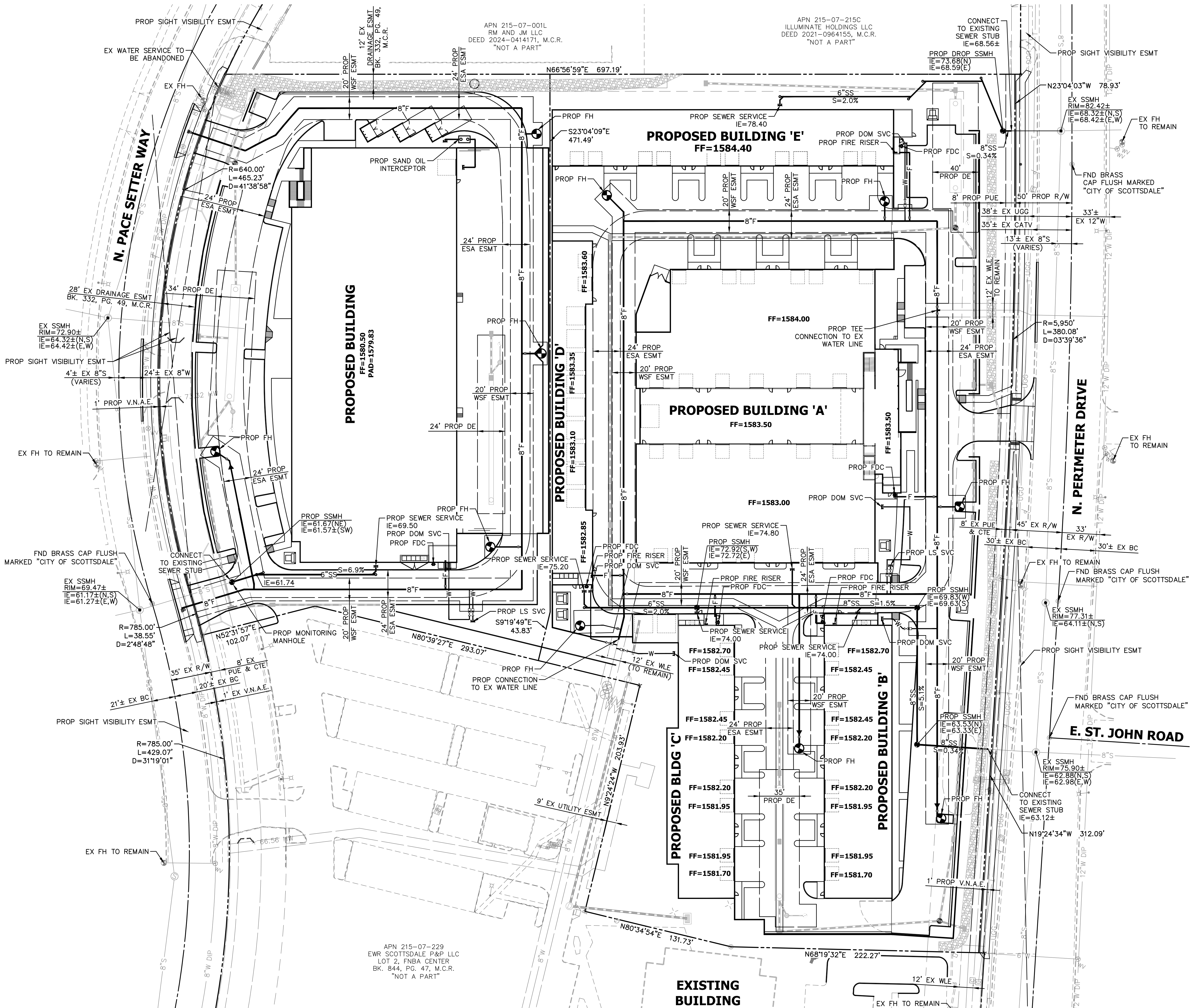
## Appendix A

### Vicinity Map and Conceptual Utility Plan



**VICINITY MAP  
FIGURE 1**



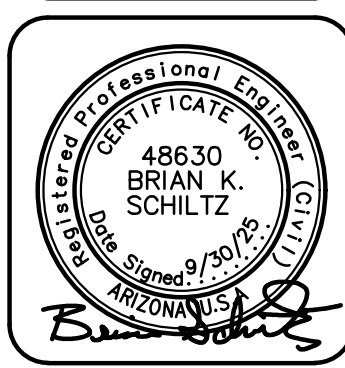


| NO. | DATE | REVISION | BY |
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|     |      |          |    |
|     |      |          |    |
|     |      |          |    |

PURPOSE: ZONING SUBMITTAL

DESIGN BY: BKS  
DRAWN BY: MM  
CHECKED BY: BKS

CIVIL AND SURVEY  
**HUNTER**  
ENGINEERING  
10446 NORTH 74TH STREET  
SUITE 140  
SCOTTSDALE, AZ 85258  
P 480 991 3985



CONCEPTUAL UTILITY PLAN  
FOR  
FINISH LINE AUTO CLUB GARAGES  
PRINCESS DRIVE AND LOOP 101  
SCOTTSDALE, ARIZONA



THESE PLANS ARE NOT APPROVED FOR CONSTRUCTION WITHOUT AN APPROVED SIGNATURE FROM THE GOVERNING MUNICIPALITY.

PROJECT NAME:  
FINISH LINE  
AUTO CLUB  
GARAGES

HE NO.: LGEC332  
SCALE: 1"=40'

SHEET:  
**C3**



**Appendix B**  
**Design Data and Calculations**

# WATER FLOW TEST REPORT

**Project:** Finish Line - Perimeter Center  
**Project Number:** LGEC332

TOTAL FLOW DURING TEST: 2946 GPM  
STATIC READING: 72 PSI      RESIDUAL: 60 PSI  
RESULTS: AT 20 PSI RESIDUAL 6503 GPM      AT 0 PSI 7752 GPM  
REMARKS:

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|                 | Shutoff Head     | Design           | Max Operating    |
|-----------------|------------------|------------------|------------------|
|                 | <u>72</u> psi    | <u>60</u> psi    | <u>0</u> psi     |
| psi X 2.31 = ft | <u>166.32</u> ft | <u>138.6</u> ft  | <u>0</u> ft      |
|                 |                  | <u>2946</u> flow | <u>7752</u> flow |

Project: Finish Line Auto Club - Perimeter Center  
Project Number: LGEC332  
City: Scottsdale  
Date: 9/29/2025

#### SITE DOMESTIC DEMAND SUMMARY

| <b>Finish Line Auto Club<br/>(Perimeter Drive)</b> | <b>Site Use</b> | <b>Building Area<br/>(sf)</b> | <b>Average Day Demand (gpm)<br/>Gross Bldg Area (sf)</b><br>per Table 6-1.2 Average Day Water Demands | <b>Average Day Demand (ADD)<br/>(gpm)</b> | <b>Average Day Demand (ADD)<br/>(gpd)</b> | <b>Max Day Demand (MDD)<br/>(ADD*2.0)<br/>(gpm)</b> | <b>Peak Hour Demand (PHD)<br/>(ADD*3.5)<br/>(gpm)</b> |
|----------------------------------------------------|-----------------|-------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------|-----------------------------------------------------|-------------------------------------------------------|
| Building A (Clubhouse)                             | Commercial      | 63,020                        | 0.001110 gpm per sf                                                                                   | 69.95                                     | 50,416                                    | 139.90                                              | 244.83                                                |
| Building B                                         | Commercial      | 17,877                        | 0.001110 gpm per sf                                                                                   | 19.84                                     | 14,302                                    | 39.69                                               | 69.45                                                 |
| Building C                                         | Commercial      | 17,337                        | 0.001110 gpm per sf                                                                                   | 19.24                                     | 13,870                                    | 38.49                                               | 67.35                                                 |
| Building D                                         | Commercial      | 8,712                         | 0.001110 gpm per sf                                                                                   | 9.67                                      | 6,970                                     | 19.34                                               | 33.85                                                 |
| Building E                                         | Commercial      | 18,558                        | 0.001110 gpm per sf                                                                                   | 20.60                                     | 14,846                                    | 41.20                                               | 72.10                                                 |
| <b>Subtotal</b>                                    |                 | <b>125,504</b>                |                                                                                                       | <b>139.31</b>                             | <b>100,403</b>                            | <b>278.62</b>                                       | <b>487.58</b>                                         |

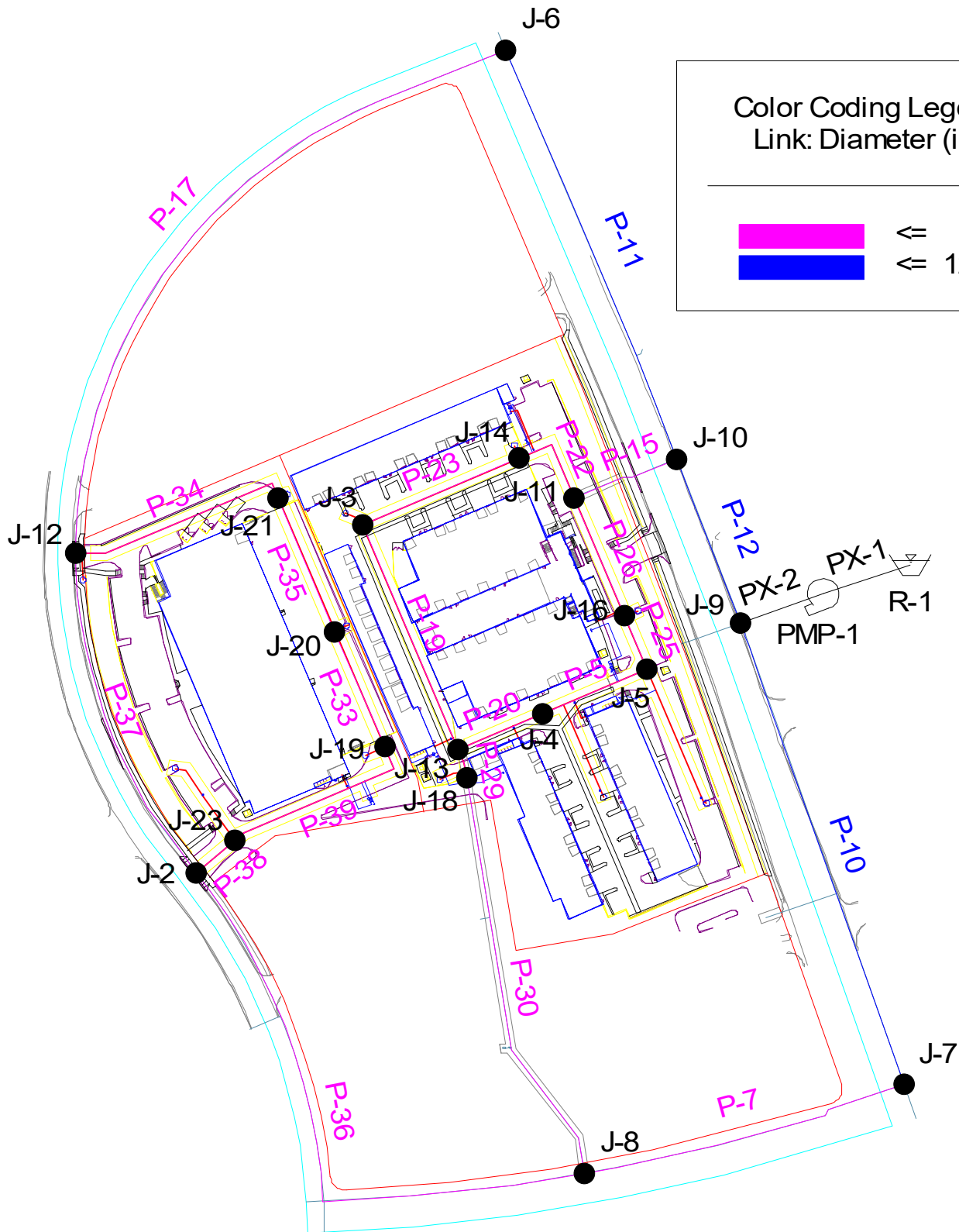
| <b>Car Storage Building<br/>(Pacesetter Way)</b> | <b>Site Use</b> | <b>Site Area (ac)</b> | <b>Average Day Demand (gpm)<br/>Site Area (ac)</b><br>per Table 6-1.2 Average Day Water Demands | <b>Average Day Demand (ADD)<br/>(gpm)</b> | <b>Average Day Demand (ADD)<br/>(gpd)</b> | <b>Max Day Demand (MDD)<br/>(ADD*2.0)<br/>(gpm)</b> | <b>Peak Hour Demand (PHD)<br/>(ADD*3.5)<br/>(gpm)</b> |
|--------------------------------------------------|-----------------|-----------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------|-----------------------------------------------------|-------------------------------------------------------|
| Car Storage Building                             | Industrial      | 3.65                  | 1.44 gpm per ac                                                                                 | 5.26                                      | 3,749                                     | 10.51                                               | 18.40                                                 |
| <b>Subtotal</b>                                  |                 | <b>3.65</b>           |                                                                                                 | <b>5.26</b>                               | <b>3,749</b>                              | <b>10.51</b>                                        | <b>18.40</b>                                          |

|                    |  |  |  |               |                |               |               |
|--------------------|--|--|--|---------------|----------------|---------------|---------------|
| <b>Grand Total</b> |  |  |  | <b>144.57</b> | <b>104,152</b> | <b>289.13</b> | <b>505.98</b> |
|--------------------|--|--|--|---------------|----------------|---------------|---------------|

#### SITE FIRE FLOW SUMMARY

| <b>Building</b>        | <b>Construction Type</b> | <b>Building Area (sf)</b> | <b>Minimum Required Fire Flow for Buildings<br/>Table B105.1</b><br>International Fire Code | <b>Min Fire Flow 25% of Table Value (gpm) or<br/>Min 1,500 gpm</b> |
|------------------------|--------------------------|---------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Building A (Clubhouse) | V-B                      | 63,020                    | 6,750 gpm                                                                                   | 1,688                                                              |
| Building B             | V-B                      | 17,877                    | 3,500 gpm                                                                                   | 1,500                                                              |
| Building C             | V-B                      | 17,337                    | 3,500 gpm                                                                                   | 1,500                                                              |
| Building D             | V-B                      | 8,712                     | 2,500 gpm                                                                                   | 1,500                                                              |
| Building E             | V-B                      | 18,558                    | 3,750 gpm                                                                                   | 1,500                                                              |
| Car Storage Building   | V-B                      | 49,266                    | 6,000 gpm                                                                                   | 1,500                                                              |

# Scenario: Finish Line - Perimeter Center



## Color Coding Legend Link: Diameter (in)

|  |        |
|--|--------|
|  | ≤ 8.0  |
|  | ≤ 12.0 |

# Scenario: Average Day Steady State Analysis Junction Report

| Label | Elevation<br>(ft) | Base Flow<br>(gpm) | Calculated<br>Hydraulic Grade<br>(ft) | Pressure<br>(psi) |
|-------|-------------------|--------------------|---------------------------------------|-------------------|
| J-3   | 1,583.55          | 0.00               | 1,743.92                              | 69.38             |
| J-4   | 1,582.57          | 0.00               | 1,743.91                              | 69.80             |
| J-5   | 1,581.75          | 89.80              | 1,743.91                              | 70.16             |
| J-6   | 1,588.10          | 0.00               | 1,743.96                              | 67.43             |
| J-7   | 1,569.80          | 0.00               | 1,743.96                              | 75.35             |
| J-8   | 1,567.50          | 0.00               | 1,743.95                              | 76.34             |
| J-9   | 1,577.75          | 0.00               | 1,743.97                              | 71.91             |
| J-10  | 1,580.60          | 0.00               | 1,743.96                              | 70.68             |
| J-11  | 1,581.50          | 0.00               | 1,743.92                              | 70.27             |
| J-12  | 1,575.33          | 0.00               | 1,743.95                              | 72.95             |
| J-13  | 1,582.70          | 0.00               | 1,743.91                              | 69.75             |
| J-16  | 1,581.20          | 0.00               | 1,743.91                              | 70.40             |
| J-18  | 1,579.00          | 28.91              | 1,743.91                              | 71.35             |
| J-19  | 1,575.40          | 5.26               | 1,743.95                              | 72.92             |
| J-20  | 1,579.00          | 0.00               | 1,743.95                              | 71.37             |
| J-21  | 1,581.40          | 0.00               | 1,743.95                              | 70.33             |
| J-2   | 1,569.25          | 0.00               | 1,743.95                              | 75.58             |
| J-14  | 1,583.77          | 20.60              | 1,743.92                              | 69.29             |
| J-23  | 1,571.45          | 0.00               | 1,743.95                              | 74.63             |

**Scenario: Average Day  
Steady State Analysis  
Pipe Report**

| Label | Start Node | Stop Node | Length (ft) | Diameter (in) | Material     | Hazen-Williams C | Discharge (gpm) | Upstream Structure Hydraulic Grade (ft) | Downstream Structure Hydraulic Grade (ft) | Pressure Pipe Headloss (ft) | Headloss Gradient (ft/1000ft) | Velocity (ft/s) |
|-------|------------|-----------|-------------|---------------|--------------|------------------|-----------------|-----------------------------------------|-------------------------------------------|-----------------------------|-------------------------------|-----------------|
| P-5   | J-4        | J-5       | 152.00      | 8.0           | PVC          | 150.0            | 34.19           | 1,743.91                                | 1,743.91                                  | 0.00                        | 0.03                          | 0.22            |
| P-7   | J-7        | J-8       | 446.00      | 8.0           | Ductile Iron | 130.0            | 35.66           | 1,743.96                                | 1,743.95                                  | 0.02                        | 0.04                          | 0.23            |
| P-10  | J-9        | J-7       | 654.00      | 12.0          | Ductile Iron | 130.0            | 35.66           | 1,743.97                                | 1,743.96                                  | 0.00                        | 0.01                          | 0.10            |
| PX-1  | R-1        | PMP-1     | 1.00        | 120.0         | Ductile Iron | 130.0            | 144.57          | 1,577.75                                | 1,577.75                                  | 0.00                        | 0.00                          | 0.00            |
| PX-2  | PMP-1      | J-9       | 1.00        | 120.0         | Ductile Iron | 130.0            | 144.57          | 1,743.97                                | 1,743.97                                  | 0.00                        | 0.00                          | 0.00            |
| P-11  | J-6        | J-10      | 593.00      | 12.0          | Ductile Iron | 130.0            | -14.17          | 1,743.96                                | 1,743.96                                  | 0.00                        | 0.00                          | 0.04            |
| P-12  | J-10       | J-9       | 237.00      | 12.0          | Ductile Iron | 130.0            | -108.91         | 1,743.96                                | 1,743.97                                  | 0.01                        | 0.04                          | 0.31            |
| P-15  | J-11       | J-10      | 147.00      | 8.0           | Ductile Iron | 130.0            | -94.74          | 1,743.92                                | 1,743.96                                  | 0.03                        | 0.23                          | 0.60            |
| P-17  | J-12       | J-6       | 957.00      | 8.0           | Ductile Iron | 130.0            | -14.17          | 1,743.95                                | 1,743.96                                  | 0.01                        | 0.01                          | 0.09            |
| P-19  | J-3        | J-13      | 327.00      | 8.0           | PVC          | 150.0            | 18.53           | 1,743.92                                | 1,743.91                                  | 0.00                        | 0.01                          | 0.12            |
| P-20  | J-13       | J-4       | 123.00      | 8.0           | PVC          | 150.0            | 34.19           | 1,743.91                                | 1,743.91                                  | 0.00                        | 0.03                          | 0.22            |
| P-25  | J-5        | J-16      | 77.00       | 8.0           | PVC          | 150.0            | -55.61          | 1,743.91                                | 1,743.91                                  | 0.01                        | 0.07                          | 0.35            |
| P-26  | J-16       | J-11      | 172.00      | 8.0           | PVC          | 150.0            | -55.61          | 1,743.91                                | 1,743.92                                  | 0.01                        | 0.07                          | 0.35            |
| P-29  | J-13       | J-18      | 41.00       | 8.0           | PVC          | 150.0            | -15.66          | 1,743.91                                | 1,743.91                                  | 0.00                        | 0.01                          | 0.10            |
| P-30  | J-18       | J-8       | 565.00      | 8.0           | Ductile Iron | 130.0            | -44.57          | 1,743.91                                | 1,743.95                                  | 0.03                        | 0.06                          | 0.28            |
| P-34  | J-12       | J-21      | 304.00      | 8.0           | PVC          | 150.0            | 6.84            | 1,743.95                                | 1,743.95                                  | 0.00                        | 0.00                          | 0.04            |
| P-33  | J-20       | J-19      | 168.00      | 8.0           | PVC          | 150.0            | 6.84            | 1,743.95                                | 1,743.95                                  | 0.00                        | 0.00                          | 0.04            |
| P-35  | J-21       | J-20      | 194.00      | 8.0           | PVC          | 150.0            | 6.84            | 1,743.95                                | 1,743.95                                  | 0.00                        | 0.00                          | 0.04            |
| P-36  | J-8        | J-2       | 832.00      | 8.0           | Ductile Iron | 130.0            | -8.91           | 1,743.95                                | 1,743.95                                  | 0.00                        | 0.00                          | 0.06            |
| P-37  | J-2        | J-12      | 465.00      | 8.0           | Ductile Iron | 130.0            | -7.33           | 1,743.95                                | 1,743.95                                  | 0.00                        | 0.00                          | 0.05            |
| P-23  | J-3        | J-14      | 229.00      | 8.0           | PVC          | 150.0            | -18.53          | 1,743.92                                | 1,743.92                                  | 0.00                        | 0.01                          | 0.12            |
| P-22  | J-14       | J-11      | 124.00      | 8.0           | PVC          | 150.0            | -39.13          | 1,743.92                                | 1,743.92                                  | 0.00                        | 0.03                          | 0.25            |
| P-38  | J-2        | J-23      | 67.00       | 8.0           | PVC          | 150.0            | -1.58           | 1,743.95                                | 1,743.95                                  | 0.00                        | 0.00                          | 0.01            |
| P-39  | J-23       | J-19      | 266.00      | 8.0           | PVC          | 150.0            | -1.58           | 1,743.95                                | 1,743.95                                  | 0.00                        | 0.00                          | 0.01            |

# Scenario: Max Day Steady State Analysis Junction Report

| Label | Elevation<br>(ft) | Base Flow<br>(gpm) | Calculated<br>Hydraulic Grade<br>(ft) | Pressure<br>(psi) |
|-------|-------------------|--------------------|---------------------------------------|-------------------|
| J-3   | 1,583.55          | 0.00               | 1,743.51                              | 69.21             |
| J-4   | 1,582.57          | 0.00               | 1,743.49                              | 69.62             |
| J-5   | 1,581.75          | 179.59             | 1,743.47                              | 69.97             |
| J-6   | 1,588.10          | 0.00               | 1,743.66                              | 67.30             |
| J-7   | 1,569.80          | 0.00               | 1,743.68                              | 75.23             |
| J-8   | 1,567.50          | 0.00               | 1,743.62                              | 76.20             |
| J-9   | 1,577.75          | 0.00               | 1,743.69                              | 71.80             |
| J-10  | 1,580.60          | 0.00               | 1,743.66                              | 70.55             |
| J-11  | 1,581.50          | 0.00               | 1,743.53                              | 70.10             |
| J-12  | 1,575.33          | 0.00               | 1,743.63                              | 72.82             |
| J-13  | 1,582.70          | 0.00               | 1,743.50                              | 69.57             |
| J-16  | 1,581.20          | 0.00               | 1,743.49                              | 70.22             |
| J-18  | 1,579.00          | 57.83              | 1,743.50                              | 71.17             |
| J-19  | 1,575.40          | 10.51              | 1,743.63                              | 72.78             |
| J-20  | 1,579.00          | 0.00               | 1,743.63                              | 71.23             |
| J-21  | 1,581.40          | 0.00               | 1,743.63                              | 70.19             |
| J-2   | 1,569.25          | 0.00               | 1,743.63                              | 75.45             |
| J-14  | 1,583.77          | 41.20              | 1,743.52                              | 69.12             |
| J-23  | 1,571.45          | 0.00               | 1,743.63                              | 74.49             |

**Scenario: Max Day**  
**Steady State Analysis**  
**Pipe Report**

| Label | Start Node | Stop Node | Length (ft) | Diameter (in) | Material     | Hazen-Williams C | Discharge (gpm) | Upstream Structure Hydraulic Grade (ft) | Downstream Structure Hydraulic Grade (ft) | Pressure Pipe Headloss (ft) | Headloss Gradient (ft/1000ft) | Velocity (ft/s) |
|-------|------------|-----------|-------------|---------------|--------------|------------------|-----------------|-----------------------------------------|-------------------------------------------|-----------------------------|-------------------------------|-----------------|
| P-5   | J-4        | J-5       | 152.00      | 8.0           | PVC          | 150.0            | 68.37           | 1,743.49                                | 1,743.47                                  | 0.01                        | 0.10                          | 0.44            |
| P-7   | J-7        | J-8       | 446.00      | 8.0           | Ductile Iron | 130.0            | 71.32           | 1,743.68                                | 1,743.62                                  | 0.06                        | 0.14                          | 0.46            |
| P-10  | J-9        | J-7       | 654.00      | 12.0          | Ductile Iron | 130.0            | 71.32           | 1,743.69                                | 1,743.68                                  | 0.01                        | 0.02                          | 0.20            |
| PX-1  | R-1        | PMP-1     | 1.00        | 120.0         | Ductile Iron | 130.0            | 289.13          | 1,577.75                                | 1,577.75                                  | 0.00                        | 0.00                          | 0.01            |
| PX-2  | PMP-1      | J-9       | 1.00        | 120.0         | Ductile Iron | 130.0            | 289.13          | 1,743.69                                | 1,743.69                                  | 0.00                        | 0.00                          | 0.01            |
| P-11  | J-6        | J-10      | 593.00      | 12.0          | Ductile Iron | 130.0            | -28.34          | 1,743.66                                | 1,743.66                                  | 0.00                        | 0.00                          | 0.08            |
| P-12  | J-10       | J-9       | 237.00      | 12.0          | Ductile Iron | 130.0            | -217.81         | 1,743.66                                | 1,743.69                                  | 0.04                        | 0.15                          | 0.62            |
| P-15  | J-11       | J-10      | 147.00      | 8.0           | Ductile Iron | 130.0            | -189.47         | 1,743.53                                | 1,743.66                                  | 0.12                        | 0.84                          | 1.21            |
| P-17  | J-12       | J-6       | 957.00      | 8.0           | Ductile Iron | 130.0            | -28.34          | 1,743.63                                | 1,743.66                                  | 0.02                        | 0.02                          | 0.18            |
| P-19  | J-3        | J-13      | 327.00      | 8.0           | PVC          | 150.0            | 37.06           | 1,743.51                                | 1,743.50                                  | 0.01                        | 0.03                          | 0.24            |
| P-20  | J-13       | J-4       | 123.00      | 8.0           | PVC          | 150.0            | 68.37           | 1,743.50                                | 1,743.49                                  | 0.01                        | 0.10                          | 0.44            |
| P-25  | J-5        | J-16      | 77.00       | 8.0           | PVC          | 150.0            | -111.22         | 1,743.47                                | 1,743.49                                  | 0.02                        | 0.24                          | 0.71            |
| P-26  | J-16       | J-11      | 172.00      | 8.0           | PVC          | 150.0            | -111.22         | 1,743.49                                | 1,743.53                                  | 0.04                        | 0.24                          | 0.71            |
| P-29  | J-13       | J-18      | 41.00       | 8.0           | PVC          | 150.0            | -31.32          | 1,743.50                                | 1,743.50                                  | 0.00                        | 0.02                          | 0.20            |
| P-30  | J-18       | J-8       | 565.00      | 8.0           | Ductile Iron | 130.0            | -89.15          | 1,743.50                                | 1,743.62                                  | 0.12                        | 0.21                          | 0.57            |
| P-34  | J-12       | J-21      | 304.00      | 8.0           | PVC          | 150.0            | 13.68           | 1,743.63                                | 1,743.63                                  | 0.00                        | 0.00                          | 0.09            |
| P-33  | J-20       | J-19      | 168.00      | 8.0           | PVC          | 150.0            | 13.68           | 1,743.63                                | 1,743.63                                  | 0.00                        | 0.01                          | 0.09            |
| P-35  | J-21       | J-20      | 194.00      | 8.0           | PVC          | 150.0            | 13.68           | 1,743.63                                | 1,743.63                                  | 0.00                        | 0.01                          | 0.09            |
| P-36  | J-8        | J-2       | 832.00      | 8.0           | Ductile Iron | 130.0            | -17.83          | 1,743.62                                | 1,743.63                                  | 0.01                        | 0.01                          | 0.11            |
| P-37  | J-2        | J-12      | 465.00      | 8.0           | Ductile Iron | 130.0            | -14.65          | 1,743.63                                | 1,743.63                                  | 0.00                        | 0.01                          | 0.09            |
| P-23  | J-3        | J-14      | 229.00      | 8.0           | PVC          | 150.0            | -37.06          | 1,743.51                                | 1,743.52                                  | 0.01                        | 0.03                          | 0.24            |
| P-22  | J-14       | J-11      | 124.00      | 8.0           | PVC          | 150.0            | -78.26          | 1,743.52                                | 1,743.53                                  | 0.02                        | 0.13                          | 0.50            |
| P-38  | J-2        | J-23      | 67.00       | 8.0           | PVC          | 150.0            | -3.17           | 1,743.63                                | 1,743.63                                  | 0.00                        | 0.00                          | 0.02            |
| P-39  | J-23       | J-19      | 266.00      | 8.0           | PVC          | 150.0            | -3.17           | 1,743.63                                | 1,743.63                                  | 0.00                        | 0.00                          | 0.02            |



# Scenario: Peak Hour Steady State Analysis Junction Report

| Label | Elevation<br>(ft) | Base Flow<br>(gpm) | Calculated<br>Hydraulic Grade<br>(ft) | Pressure<br>(psi) |
|-------|-------------------|--------------------|---------------------------------------|-------------------|
| J-3   | 1,583.55          | 0.00               | 1,742.50                              | 68.77             |
| J-4   | 1,582.57          | 0.00               | 1,742.43                              | 69.17             |
| J-5   | 1,581.75          | 314.28             | 1,742.39                              | 69.50             |
| J-6   | 1,588.10          | 0.00               | 1,742.90                              | 66.98             |
| J-7   | 1,569.80          | 0.00               | 1,742.97                              | 74.92             |
| J-8   | 1,567.50          | 0.00               | 1,742.80                              | 75.84             |
| J-9   | 1,577.75          | 0.00               | 1,743.01                              | 71.50             |
| J-10  | 1,580.60          | 0.00               | 1,742.91                              | 70.22             |
| J-11  | 1,581.50          | 0.00               | 1,742.56                              | 69.68             |
| J-12  | 1,575.33          | 0.00               | 1,742.84                              | 72.47             |
| J-13  | 1,582.70          | 0.00               | 1,742.47                              | 69.12             |
| J-16  | 1,581.20          | 0.00               | 1,742.44                              | 69.76             |
| J-18  | 1,579.00          | 101.20             | 1,742.47                              | 70.73             |
| J-19  | 1,575.40          | 18.40              | 1,742.83                              | 72.44             |
| J-20  | 1,579.00          | 0.00               | 1,742.83                              | 70.88             |
| J-21  | 1,581.40          | 0.00               | 1,742.83                              | 69.84             |
| J-2   | 1,569.25          | 0.00               | 1,742.83                              | 75.10             |
| J-14  | 1,583.77          | 72.10              | 1,742.52                              | 68.68             |
| J-23  | 1,571.45          | 0.00               | 1,742.83                              | 74.15             |

**Scenario: Peak Hour  
Steady State Analysis  
Pipe Report**

| Label | Start Node | Stop Node | Length (ft) | Diameter (in) | Material     | Hazen-Williams C | Discharge (gpm) | Upstream Structure Hydraulic Grade (ft) | Downstream Structure Hydraulic Grade (ft) | Pressure Pipe Headloss (ft) | Headloss Gradient (ft/1000ft) | Velocity (ft/s) |
|-------|------------|-----------|-------------|---------------|--------------|------------------|-----------------|-----------------------------------------|-------------------------------------------|-----------------------------|-------------------------------|-----------------|
| P-5   | J-4        | J-5       | 152.00      | 8.0           | PVC          | 150.0            | 119.65          | 1,742.43                                | 1,742.39                                  | 0.04                        | 0.27                          | 0.76            |
| P-7   | J-7        | J-8       | 446.00      | 8.0           | Ductile Iron | 130.0            | 124.81          | 1,742.97                                | 1,742.80                                  | 0.17                        | 0.39                          | 0.80            |
| P-10  | J-9        | J-7       | 654.00      | 12.0          | Ductile Iron | 130.0            | 124.81          | 1,743.01                                | 1,742.97                                  | 0.04                        | 0.05                          | 0.35            |
| PX-1  | R-1        | PMP-1     | 1.00        | 120.0         | Ductile Iron | 130.0            | 505.98          | 1,577.75                                | 1,577.75                                  | 0.00                        | 0.00                          | 0.01            |
| PX-2  | PMP-1      | J-9       | 1.00        | 120.0         | Ductile Iron | 130.0            | 505.98          | 1,743.01                                | 1,743.01                                  | 0.00                        | 0.00                          | 0.01            |
| P-11  | J-6        | J-10      | 593.00      | 12.0          | Ductile Iron | 130.0            | -49.59          | 1,742.90                                | 1,742.91                                  | 0.01                        | 0.01                          | 0.14            |
| P-12  | J-10       | J-9       | 237.00      | 12.0          | Ductile Iron | 130.0            | -381.17         | 1,742.91                                | 1,743.01                                  | 0.10                        | 0.42                          | 1.08            |
| P-15  | J-11       | J-10      | 147.00      | 8.0           | Ductile Iron | 130.0            | -331.58         | 1,742.56                                | 1,742.91                                  | 0.35                        | 2.37                          | 2.12            |
| P-17  | J-12       | J-6       | 957.00      | 8.0           | Ductile Iron | 130.0            | -49.59          | 1,742.84                                | 1,742.90                                  | 0.07                        | 0.07                          | 0.32            |
| P-19  | J-3        | J-13      | 327.00      | 8.0           | PVC          | 150.0            | 64.85           | 1,742.50                                | 1,742.47                                  | 0.03                        | 0.09                          | 0.41            |
| P-20  | J-13       | J-4       | 123.00      | 8.0           | PVC          | 150.0            | 119.65          | 1,742.47                                | 1,742.43                                  | 0.03                        | 0.27                          | 0.76            |
| P-25  | J-5        | J-16      | 77.00       | 8.0           | PVC          | 150.0            | -194.63         | 1,742.39                                | 1,742.44                                  | 0.05                        | 0.68                          | 1.24            |
| P-26  | J-16       | J-11      | 172.00      | 8.0           | PVC          | 150.0            | -194.63         | 1,742.44                                | 1,742.56                                  | 0.12                        | 0.68                          | 1.24            |
| P-29  | J-13       | J-18      | 41.00       | 8.0           | PVC          | 150.0            | -54.80          | 1,742.47                                | 1,742.47                                  | 0.00                        | 0.06                          | 0.35            |
| P-30  | J-18       | J-8       | 565.00      | 8.0           | Ductile Iron | 130.0            | -156.00         | 1,742.47                                | 1,742.80                                  | 0.33                        | 0.59                          | 1.00            |
| P-34  | J-12       | J-21      | 304.00      | 8.0           | PVC          | 150.0            | 23.95           | 1,742.84                                | 1,742.83                                  | 0.00                        | 0.01                          | 0.15            |
| P-33  | J-20       | J-19      | 168.00      | 8.0           | PVC          | 150.0            | 23.95           | 1,742.83                                | 1,742.83                                  | 0.00                        | 0.01                          | 0.15            |
| P-35  | J-21       | J-20      | 194.00      | 8.0           | PVC          | 150.0            | 23.95           | 1,742.83                                | 1,742.83                                  | 0.00                        | 0.01                          | 0.15            |
| P-36  | J-8        | J-2       | 832.00      | 8.0           | Ductile Iron | 130.0            | -31.19          | 1,742.80                                | 1,742.83                                  | 0.02                        | 0.03                          | 0.20            |
| P-37  | J-2        | J-12      | 465.00      | 8.0           | Ductile Iron | 130.0            | -25.65          | 1,742.83                                | 1,742.84                                  | 0.01                        | 0.02                          | 0.16            |
| P-23  | J-3        | J-14      | 229.00      | 8.0           | PVC          | 150.0            | -64.85          | 1,742.50                                | 1,742.52                                  | 0.02                        | 0.09                          | 0.41            |
| P-22  | J-14       | J-11      | 124.00      | 8.0           | PVC          | 150.0            | -136.95         | 1,742.52                                | 1,742.56                                  | 0.04                        | 0.35                          | 0.87            |
| P-38  | J-2        | J-23      | 67.00       | 8.0           | PVC          | 150.0            | -5.55           | 1,742.83                                | 1,742.83                                  | 0.00                        | 0.00                          | 0.04            |
| P-39  | J-23       | J-19      | 266.00      | 8.0           | PVC          | 150.0            | -5.55           | 1,742.83                                | 1,742.83                                  | 0.00                        | 0.00                          | 0.04            |

# Scenario: Fire+Max Day - Worst-Case Scenario

## Steady State Analysis

### Junction Report

| Label | Elevation (ft) | Base Flow (gpm) | Calculated Hydraulic Grade (ft) | Pressure (psi) |
|-------|----------------|-----------------|---------------------------------|----------------|
| J-3   | 1,583.55       | 0.00            | 1,731.10                        | 63.84          |
| J-4   | 1,582.57       | 0.00            | 1,730.94                        | 64.19          |
| J-5   | 1,581.75       | 179.59          | 1,730.98                        | 64.56          |
| J-6   | 1,588.10       | 0.00            | 1,731.25                        | 61.93          |
| J-7   | 1,569.80       | 0.00            | 1,732.57                        | 70.42          |
| J-8   | 1,567.50       | 0.00            | 1,730.12                        | 70.36          |
| J-9   | 1,577.75       | 0.00            | 1,733.06                        | 67.20          |
| J-10  | 1,580.60       | 0.00            | 1,732.13                        | 65.56          |
| J-11  | 1,581.50       | 0.00            | 1,731.34                        | 64.83          |
| J-12  | 1,575.33       | 0.00            | 1,720.98                        | 63.01          |
| J-13  | 1,582.70       | 0.00            | 1,730.91                        | 64.12          |
| J-16  | 1,581.20       | 0.00            | 1,731.09                        | 64.85          |
| J-18  | 1,579.00       | 57.83           | 1,730.85                        | 65.70          |
| J-19  | 1,575.40       | 10.51           | 1,718.20                        | 61.78          |
| J-20  | 1,579.00       | 1,500.00        | 1,716.83                        | 59.63          |
| J-21  | 1,581.40       | 0.00            | 1,718.45                        | 59.29          |
| J-2   | 1,569.25       | 0.00            | 1,720.98                        | 65.65          |
| J-14  | 1,583.77       | 41.20           | 1,731.23                        | 63.80          |
| J-23  | 1,571.45       | 0.00            | 1,720.42                        | 64.45          |

Bldg A + Bldg B = 139.90 + 39.69 = 179.59 gpm

Bldg C + Bldg D = 38.49 + 19.34 = 57.83 gpm

**Scenario: Fire+Max Day - Worst-Case Scenario**  
**Steady State Analysis**  
**Pipe Report**

| Label | Start Node | Stop Node | Length (ft) | Diameter (in) | Material     | Hazen-Williams C | Discharge (gpm) | Upstream Structure Hydraulic Grade (ft) | Downstream Structure Hydraulic Grade (ft) | Pressure Pipe Headloss (ft) | Headloss Gradient (ft/1000ft) | Velocity (ft/s) |
|-------|------------|-----------|-------------|---------------|--------------|------------------|-----------------|-----------------------------------------|-------------------------------------------|-----------------------------|-------------------------------|-----------------|
| P-5   | J-4        | J-5       | 152.00      | 8.0           | PVC          | 150.0            | -115.38         | 1,730.94                                | 1,730.98                                  | 0.04                        | 0.26                          | 0.74            |
| P-7   | J-7        | J-8       | 446.00      | 8.0           | Ductile Iron | 130.0            | 521.93          | 1,732.57                                | 1,730.12                                  | 2.44                        | 5.48                          | 3.33            |
| P-10  | J-9        | J-7       | 654.00      | 12.0          | Ductile Iron | 130.0            | 521.93          | 1,733.06                                | 1,732.57                                  | 0.50                        | 0.76                          | 1.48            |
| PX-1  | R-1        | PMP-1     | 1.00        | 120.0         | Ductile Iron | 130.0            | 1,789.13        | 1,577.75                                | 1,577.75                                  | 0.00                        | 0.00                          | 0.05            |
| PX-2  | PMP-1      | J-9       | 1.00        | 120.0         | Ductile Iron | 130.0            | 1,789.13        | 1,733.06                                | 1,733.06                                  | 0.00                        | 0.00                          | 0.05            |
| P-11  | J-6        | J-10      | 593.00      | 12.0          | Ductile Iron | 130.0            | -750.40         | 1,731.25                                | 1,732.13                                  | 0.88                        | 1.49                          | 2.13            |
| P-12  | J-10       | J-9       | 237.00      | 12.0          | Ductile Iron | 130.0            | -1,267.20       | 1,732.13                                | 1,733.06                                  | 0.93                        | 3.93                          | 3.59            |
| P-15  | J-11       | J-10      | 147.00      | 8.0           | Ductile Iron | 130.0            | -516.79         | 1,731.34                                | 1,732.13                                  | 0.79                        | 5.38                          | 3.30            |
| P-17  | J-12       | J-6       | 957.00      | 8.0           | Ductile Iron | 130.0            | -750.40         | 1,720.98                                | 1,731.25                                  | 10.27                       | 10.73                         | 4.79            |
| P-19  | J-3        | J-13      | 327.00      | 8.0           | PVC          | 150.0            | 180.63          | 1,731.10                                | 1,730.91                                  | 0.19                        | 0.59                          | 1.15            |
| P-20  | J-13       | J-4       | 123.00      | 8.0           | PVC          | 150.0            | -115.38         | 1,730.91                                | 1,730.94                                  | 0.03                        | 0.26                          | 0.74            |
| P-25  | J-5        | J-16      | 77.00       | 8.0           | PVC          | 150.0            | -294.97         | 1,730.98                                | 1,731.09                                  | 0.11                        | 1.46                          | 1.88            |
| P-26  | J-16       | J-11      | 172.00      | 8.0           | PVC          | 150.0            | -294.97         | 1,731.09                                | 1,731.34                                  | 0.25                        | 1.46                          | 1.88            |
| P-29  | J-13       | J-18      | 41.00       | 8.0           | PVC          | 150.0            | 296.00          | 1,730.91                                | 1,730.85                                  | 0.06                        | 1.47                          | 1.89            |
| P-30  | J-18       | J-8       | 565.00      | 8.0           | Ductile Iron | 130.0            | 238.17          | 1,730.85                                | 1,730.12                                  | 0.72                        | 1.28                          | 1.52            |
| P-34  | J-12       | J-21      | 304.00      | 8.0           | PVC          | 150.0            | 754.70          | 1,720.98                                | 1,718.45                                  | 2.53                        | 8.32                          | 4.82            |
| P-33  | J-20       | J-19      | 168.00      | 8.0           | PVC          | 150.0            | -745.30         | 1,716.83                                | 1,718.20                                  | 1.37                        | 8.13                          | 4.76            |
| P-35  | J-21       | J-20      | 194.00      | 8.0           | PVC          | 150.0            | 754.70          | 1,718.45                                | 1,716.83                                  | 1.61                        | 8.32                          | 4.82            |
| P-36  | J-8        | J-2       | 832.00      | 8.0           | Ductile Iron | 130.0            | 760.11          | 1,730.12                                | 1,720.98                                  | 9.14                        | 10.99                         | 4.85            |
| P-37  | J-2        | J-12      | 465.00      | 8.0           | Ductile Iron | 130.0            | 4.30            | 1,720.98                                | 1,720.98                                  | 0.00                        | 0.00                          | 0.03            |
| P-23  | J-3        | J-14      | 229.00      | 8.0           | PVC          | 150.0            | -180.63         | 1,731.10                                | 1,731.23                                  | 0.13                        | 0.59                          | 1.15            |
| P-22  | J-14       | J-11      | 124.00      | 8.0           | PVC          | 150.0            | -221.83         | 1,731.23                                | 1,731.34                                  | 0.11                        | 0.86                          | 1.42            |
| P-38  | J-2        | J-23      | 67.00       | 8.0           | PVC          | 150.0            | 755.81          | 1,720.98                                | 1,720.42                                  | 0.56                        | 8.34                          | 4.82            |
| P-39  | J-23       | J-19      | 266.00      | 8.0           | PVC          | 150.0            | 755.81          | 1,720.42                                | 1,718.20                                  | 2.22                        | 8.34                          | 4.82            |

## Detailed Report for Reservoir: R-1

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### Scenario Summary

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|                              |                           |
|------------------------------|---------------------------|
| Scenario                     | Fire+Max Day - Commercial |
| Active Topology Alternative  | Base-Active Topology      |
| Physical Alternative         | Base-Physical             |
| Demand Alternative           | Demand-Fire+Max Day       |
| Initial Settings Alternative | Base-Initial Settings     |
| Operational Alternative      | Base-Operational          |
| Age Alternative              | Base-Age Alternative      |
| Constituent Alternative      | Base-Constituent          |
| Trace Alternative            | Base-Trace Alternative    |
| Fire Flow Alternative        | Base-Fire Flow            |
| Capital Cost Alternative     | Base-Capital Cost         |
| Energy Cost Alternative      | Base-Energy Cost          |
| User Data Alternative        | Base-User Data            |

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### Global Adjustments Summary

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|        |        |           |        |
|--------|--------|-----------|--------|
| Demand | <None> | Roughness | <None> |
|--------|--------|-----------|--------|

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### Geometric Summary

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|   |               |           |             |
|---|---------------|-----------|-------------|
| X | 705,783.82 ft | Elevation | 1,577.75 ft |
| Y | 962,842.39 ft | Zone      | Zone        |

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### Calculated Results Summary

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| Time<br>(hr) | Calculated<br>Hydraulic Grade<br>(ft) | Inflow<br>(gpm) | Outflow<br>(gpm) |
|--------------|---------------------------------------|-----------------|------------------|
| 0.00         | 1,577.75                              | -1,966.62       | 1,966.62         |

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## Detailed Report for Pump: PMP-1

### Scenario Summary

|                              |                           |
|------------------------------|---------------------------|
| Scenario                     | Fire+Max Day - Commercial |
| Active Topology Alternative  | Base-Active Topology      |
| Physical Alternative         | Base-Physical             |
| Demand Alternative           | Demand-Fire+Max Day       |
| Initial Settings Alternative | Base-Initial Settings     |
| Operational Alternative      | Base-Operational          |
| Age Alternative              | Base-Age Alternative      |
| Constituent Alternative      | Base-Constituent          |
| Trace Alternative            | Base-Trace Alternative    |
| Fire Flow Alternative        | Base-Fire Flow            |
| Capital Cost Alternative     | Base-Capital Cost         |
| Energy Cost Alternative      | Base-Energy Cost          |
| User Data Alternative        | Base-User Data            |

### Global Adjustments Summary

|        |        |           |        |
|--------|--------|-----------|--------|
| Demand | <None> | Roughness | <None> |
|--------|--------|-----------|--------|

### Geometric Summary

|           |               |                 |      |
|-----------|---------------|-----------------|------|
| X         | 705,666.53 ft | Upstream Pipe   | PX-1 |
| Y         | 962,803.83 ft | Downstream Pipe | PX-2 |
| Elevation | 1,577.75 ft   |                 |      |

### Pump Definition Summary

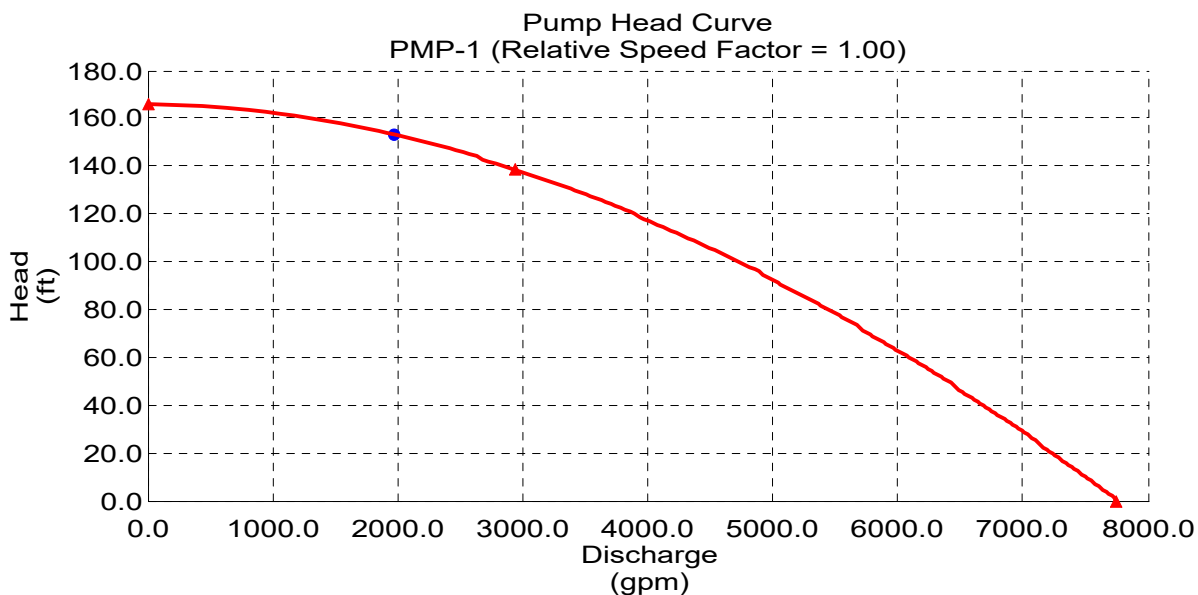
|                 |                         |
|-----------------|-------------------------|
| Pump Definition | Default Pump Definition |
|-----------------|-------------------------|

### Initial Status

|                     |    |                               |      |
|---------------------|----|-------------------------------|------|
| Initial Pump Status | On | Initial Relative Speed Factor | 1.00 |
|---------------------|----|-------------------------------|------|

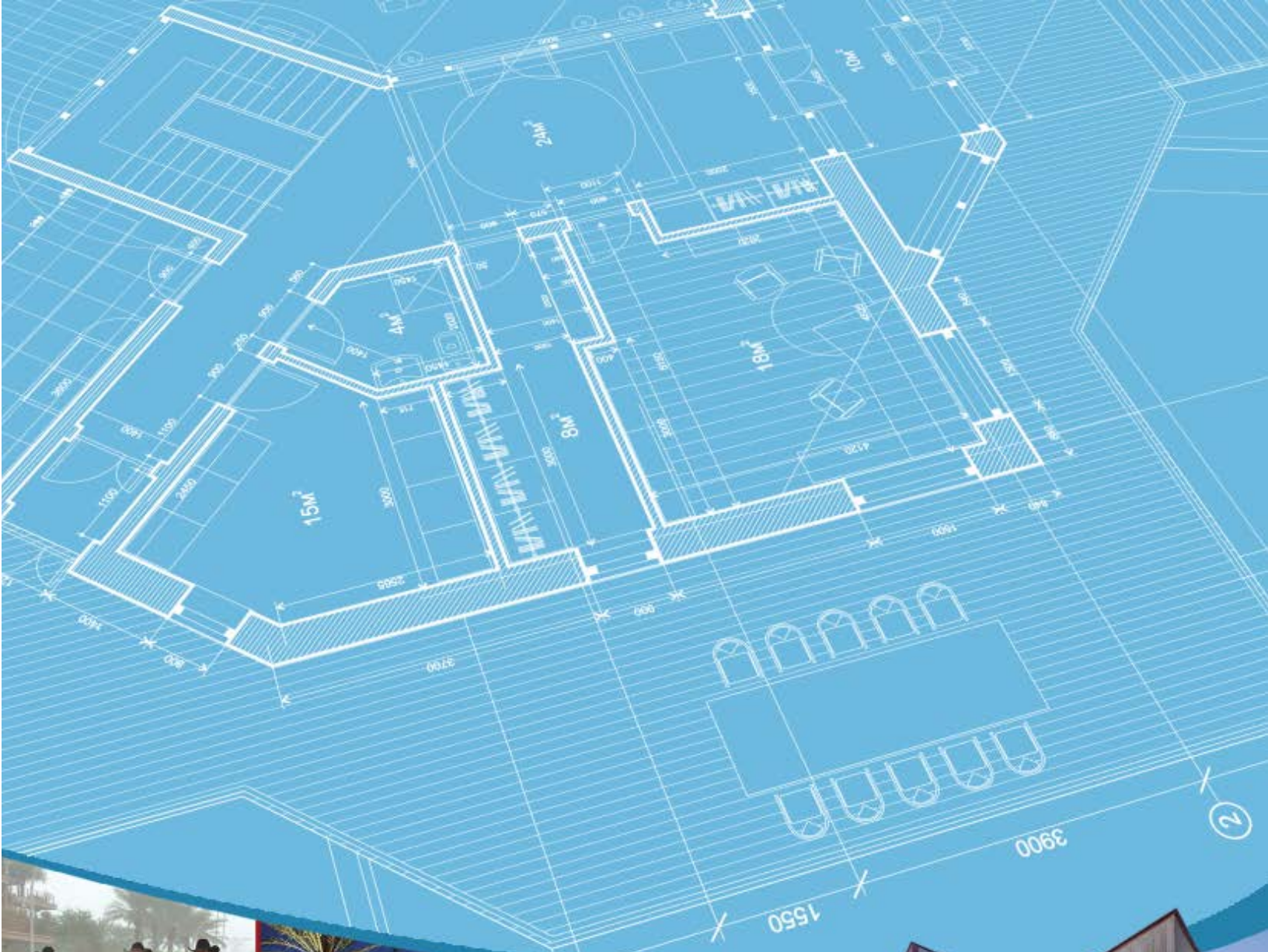
### Calculated Results Summary

| Time (hr) | Control Status | Intake Pump Grade (ft) | Discharge Pump Grade (ft) | Discharge (gpm) | Pump Head (ft) | Relative Speed | Calculated Water Power (Hp) |
|-----------|----------------|------------------------|---------------------------|-----------------|----------------|----------------|-----------------------------|
| 0.00      | On             | 1,577.75               | 1,730.96                  | 1,966.62        | 153.21         | 1.00           | 76.07                       |



**Appendix C**  
**Reference Information**





# DESIGN STANDARDS & POLICIES MANUAL



are met. The necessary process to develop the design system supply curve or grade line from the hydrant test results is described in Section 6-1.405.

3. Water demands shall generally be determined based on the unit demands in gallons per minute (gpm) listed in Figure 6-1.2, or as accepted or directed by the Water Resources Department, or as described within this chapter. The Water Resources Department reserves the right to require the use of potentially higher water demand peaking factors or directly apply demand flows for developments if deemed justified e.g. some restaurants or specialty developments.
4. Computer hydraulic modeling shall use H2ONET, WATERCAD, or EPANET software, or other Water Resource Department approved software.
5. In addition to network diagrams clearly present all inputs, details, and analysis output in organized tables.
6. Include all relevant hydraulic network diagrams listed for each of the following required hydraulic analysis scenarios:
  - a. Model Scenario 1: Average day demand in gpm at all demand nodes. (Refer to Figure 6-1.2)
  - b. Model Scenario 2: Peak hour demand in gpm at all demand nodes. (Refer to Figure 6-1.2 and Section 6-1.404 for peak hour demand)
  - c. Model Scenario 3: Maximum day demand in gpm at all demand nodes with worst case fire flow (Refer to section 6-1.501 describing fire flow determination. Refer to Figure 6-1.2 and Section 6-1.404 for maximum day demand)

Requirements:

- i. The determined fire flow must be applied to the single worst-case location in the proposed system where fire flow will be required. Typically, this is the furthest and/or highest point from the main water supply connection. If not clear what the worst-case fire flow demand point is, the fire flow shall be applied to each potential point in the model until the worst-case point is determined.
- ii. A minimum of 30 pounds per square inch (psi) must be maintained at the worst-case hydrant supply line tee/tap under this condition with a simultaneous minimum of 15 psi maintained at all domestic demand nodes (i.e., at the highest finished floor elevation and post service line and appurtenances) (Refer to scenario 4 for guidance on modeling the service line up to the demand nodes).
- d. Model Scenario 4: Maintain the minimum domestic service pressure at the worst case domestic demand node (location/elevation) under normal daily operating flow conditions termed henceforth as the *Initial Service Line Design Flow*.

Notes: A minimum of 50 psi must be maintained at the highest proposed finished floor elevation to be served, for the worst hydraulic case domestic demand node, while applying the Initial Service Line Design Flow to the node. Typically, this will be the demand node with the lowest modeled pressure in the previous scenarios. The engineer shall define the service line

details, demand node location, and determine normal daily operating hydraulic conditions/criteria as follows:

- i. Demand point location: the furthest, highest (i.e. worst case hydraulic metered node). If the service line distances and building heights are unknown, the demand node shall be located at the geometric center of the lot at elevation 12 feet above the planned finished floor elevation for single family residential. Otherwise a typical highest finished floor elevation for the development type shall be used.
- ii. Initial Service Line Design Flow: 1) Estimate the average number of water fixtures served by the subject node; 2) Use the 2015 International Plumbing Code, Appendix E, Table E103.3(2)-total load values and Table E103.3(3) to determine the normal operating flow rate; 3) Add 10 gpm minimum to the normal operating flow to account for either a hose bib or a single irrigation sprinkler zone or estimate a higher applicable intermittent or constant base flow; 4) Apply a 1.5 safety factor to the resultant total flow rate to obtain the Initial Service Line Design Flow. Note that this flow also factors into meter sizing, refer to section 6-1.416 Service Lines and Meters.
- iii. Determine the required service line and appurtenance sizing: If the sprinkler system and the domestic uses are metered through a shared meter use the greater of the resultant flow in step above, or the required fire sprinkler flow. Refer to the applicable Fire Code for sprinkler system flow and pressure requirements. Refer to section 6-1.404 Design Flows and Head Loss for design criteria on service lines.
- iv. Model pressure losses between the service tap and the demand node: Determine the losses through the water meter and the pressure reducing valve for the resultant flow and sizing from the step above. A combined 10 psi or greater loss shall be used for meter and pressure reducing valve (PRV) in scenario 4. A 5 psi or greater loss shall be used for the meter and PRV in other modeled scenarios. The service pipe friction loss portion will be per its length and diameter as included in the model.
- v. All other demand nodes in the network, other than the worst-case node shall be assigned their corresponding peak hour total use demand per Figure 6.1-2 and Section 6-1.404 during this scenario.
- vi. No fire flows are to be applied for this scenario.

#### **H. Network Diagrams**

1. Network Diagram 1: Describe the Physical Modeled Network  
Refer to Sections 6-1.300, 6-1.400, and 6-1.500 and their related subsections for water network design requirements.  
Present all the nodes, valves, pipes, tanks, hydrants, and pumps within the network and present the following:
  - a. Network components with IDs
  - b. Existing distributions system pipelines (label as existing) and connection to the proposed system

- c. Complete proposed water system, including:
  - d. Each/all individually metered service lines ending in a demand node
    - i. Irrigation connections with demand nodes (show both dedicated metered lines and where shared with the domestic service line post-meter)
    - ii. Fire sprinkler riser connections with demand nodes (typically dedicated connections are used for developments other than single family residential)
  - e. Elevations (ft.)
  - f. Pipe lengths (ft.)
  - g. Friction head loss coefficients used
  - h. Pipe/valve diameters (in)
  - i. Point of any changes in pipe diameter
  - j. Valve types and positions (open/closed/modulating)
  - k. Tanks: working volumes (gal), heights (ft.), diameters (ft.)
  - l. Pump curves (3-point min) or hydraulic grade line (HGL)
  - m. Note: Developments supplied from the city's water distribution system should be modeled as a pumped supply using a pump curve developed from the required hydrant test to simulate the dynamic flow vs. pressure supplied from the water distribution system. Alternatively, differing supply HGLs derived from the flow test results can be used for the different modeling scenarios.
  - n. Provide all valve, tank, pump, and system settings (levels, pressures/control set points, valve open/close/modulation rates and settings, valve coefficients, initial conditions, etc.)
  - o. Static system supply pressure (if supply is from system, develop pump curve or HGL from hydrant test)
- Shows all numerical units or provide a legend that indicates the units used
2. Network Diagram 2: Describe the Specific Demand Scenario
    - a. Label the figure with the title of specific modeling scenario
    - b. Indicate the demand (gpm) being applied to each applicable node for the scenario
    - c. Demand nodes shall include all applicable demands for the defined scenario including domestic use (indoor), irrigation (outdoor), fire sprinkler or hydrant, etc. If a node is a summation of more than one demand type describe this in a notes column in the associated tables
    - d. Provide all valve, tank, pump, or system settings that are specific to the modeling scenario
    - e. Shows all units or provide a legend that indicates the units used
  3. Network Diagram 3: Present the Scenario Analysis Results
 

Label the figure with the title of the specific modeling scenario and present the following output:

    - a. Node pressures (psi or HGL elevation)
    - b. Pipe flow (gpm)
    - c. Flow direction arrows

- d. Pipe flow velocity in feet per second (fps)
- e. Each pipe segment's head loss rate (ft. /1,000ft or psi/ft.)
- f. PRVs: Upstream and downstream pressures (psi or HGL elevation)
- g. Tanks: Inflow and outflow (gpm)
- h. Shows all units for the values presented or provide a legend on the diagram page that indicates the units used

| AVERAGE DAY WATER DEMANDS <sup>(1)</sup> |            |             |           |                                               |             |           |                          |
|------------------------------------------|------------|-------------|-----------|-----------------------------------------------|-------------|-----------|--------------------------|
| IN GALLONS PER DAY (GPD) <sup>(2)</sup>  |            |             |           | IN GALLONS PER MINUTE (GPM) <sup>(2)(3)</sup> |             |           |                          |
| Land Use                                 | Inside Use | Outside Use | Total Use | Inside Use                                    | Outside Use | Total Use | Units                    |
| Residential Demand per Dwelling Unit     |            |             |           |                                               |             |           |                          |
| < 2 dwelling unit per acre (DU/ac)       | 208.9      | 276.7       | 485.6     | 0.30                                          | 0.39        | 0.69      | per unit                 |
| 2 – 2.9 DU/ac                            | 193.7      | 276.7       | 470.4     | 0.27                                          | 0.39        | 0.66      | per unit                 |
| 3 – 7.9 DU/ac                            | 175.9      | 72.3        | 248.2     | 0.25                                          | 0.11        | 0.36      | per unit                 |
| 8 – 11.9 DU/ac                           | 155.3      | 72.3        | 227.6     | 0.22                                          | 0.11        | 0.33      | per unit                 |
| 12 – 22 DU/ac                            | 155.3      | 72.3        | 227.6     | 0.22                                          | 0.11        | 0.33      | per unit                 |
| High Density Condominium (condo)         | 155.3      | 30          | 185.3     | 0.22                                          | 0.05        | 0.27      | per unit                 |
| Resort Hotel (includes site amenities)   | 401.7      | 44.6        | 446.3     | 0.56                                          | 0.07        | 0.63      | per room                 |
| Service and Employment                   |            |             |           |                                               |             |           |                          |
| Restaurant                               | 1.2        | 0.1         | 1.3       | 1.67E-03                                      | 1.39E-04    | 1.81E-03  | per square foot (sq.ft.) |
| Commercial/ Retail                       | 0.7        | 0.1         | 0.8       | 9.73E-04                                      | 1.39E-04    | 1.11E-03  | per sq.ft.               |
| Commercial High Rise                     | 0.5        | 0.1         | 0.6       | 6.95E-04                                      | 1.39E-04    | 8.34E-04  | per sq.ft.               |

| AVERAGE DAY WATER DEMANDS <sup>(1)</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |      |      |                                               |          |          |            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|-----------------------------------------------|----------|----------|------------|
| IN GALLONS PER DAY (GPD) <sup>(2)</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |      |      | IN GALLONS PER MINUTE (GPM) <sup>(2)(3)</sup> |          |          |            |
| Office                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.5  | 0.1  | 0.6  | 6.95E-04                                      | 1.39E-04 | 8.34E-04 | per sq.ft. |
| Institutional                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 670  | 670  | 1340 | 0.94                                          | 0.94     | 1.88     | per acre   |
| Industrial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 873  | 154  | 1027 | 1.22                                          | 0.22     | 1.44     | per acre   |
| Research and Development                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1092 | 192  | 1284 | 1.52                                          | 0.27     | 1.79     | per acre   |
| Special Use Areas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |      |      |                                               |          |          |            |
| Natural Area Open Space                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0    | 0    | 0    | 0.0                                           | 0.0      | 0.0      | per acre   |
| Developed Open Space – Parks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0    | 1786 | 1786 | 0.0                                           | 2.49     | 2.49     | per acre   |
| Developed Open Space – Golf Course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0    | 4285 | 4285 | 0.0                                           | 5.96     | 5.96     | per acre   |
| Notes:<br>(1) These values shall not be used directly for service line or water meter sizing.<br>(2) Gallon per day values are provided for reference only. The instantaneous gallon per minute flow rates presented are intended for use in the required hydraulic modeling scenarios. The gpm values assume a 12-hour active water use period per 24-hour day. In large or specialty developments or master plans the hydraulic analysis criteria and parameters should be discussed with the Water Resources Department. Seasonal peaking should also be considered. Upon review, the Water Resources Department reserves the right to designate flows to be used in hydraulic modeling scenarios that may be different from those presented here.<br>(3) The hydraulic modeling peaking factors used in select modeling scenarios are to be applied to the gpm values shown here. Max day and peak hour peaking factors can be found in Section 6-1.404. |      |      |      |                                               |          |          |            |

FIGURE 6-1.2 AVERAGE DAY WATER DEMANDS

**Appendix D**  
**Fire Flow Test**

# Arizona Flow Testing LLC

## HYDRANT FLOW TEST REPORT

|                                     |                                                                |
|-------------------------------------|----------------------------------------------------------------|
| Project Name:                       | Finish Line - Perimeter Center                                 |
| Project Address:                    | 17600 North Perimeter Center Drive, Scottsdale, Arizona, 85255 |
| Client Project No.:                 | LGEC332                                                        |
| Arizona Flow Testing Project No.:   | 25290                                                          |
| Flow Test Permit No.:               | C78455                                                         |
| Date and time flow test conducted:  | April 14, 2025 at 6:55 AM                                      |
| Data is current and reliable until: | October 14, 2025                                               |
| Conducted by:                       | Floyd Vaughan-AZ Flow Testing, LLC (480-250-8154)              |
| Witnessed by:                       | Chris Mendez – City of Scottsdale-Inspector (602-9028-9046)    |

### Raw Test Data

Static Pressure: **90.0 PSI**  
(Measured in pounds per square inch)

Residual Pressure: **78.0 PSI**  
(Measured in pounds per square inch)

Pitot Pressure: **47.0 PSI**  
(Measured in pounds per square inch)

Diffuser Orifice Diameter: One 4-inch Pollard Diffuser  
(Measured in inches)

Coefficient of Diffuser: 0.9

Flowing GPM: **2,946 GPM**  
(Measured in gallons per minute)

GPM @ 20 PSI: **7,635 GPM**

### Data with 18 PSI Safety Factor

Static Pressure: **72.0 PSI**  
(Measured in pounds per square inch)

Residual Pressure: **60.0 PSI**  
(Measured in pounds per square inch)

Approx. distance between hydrants: 180 Feet

Main size: Not Provided

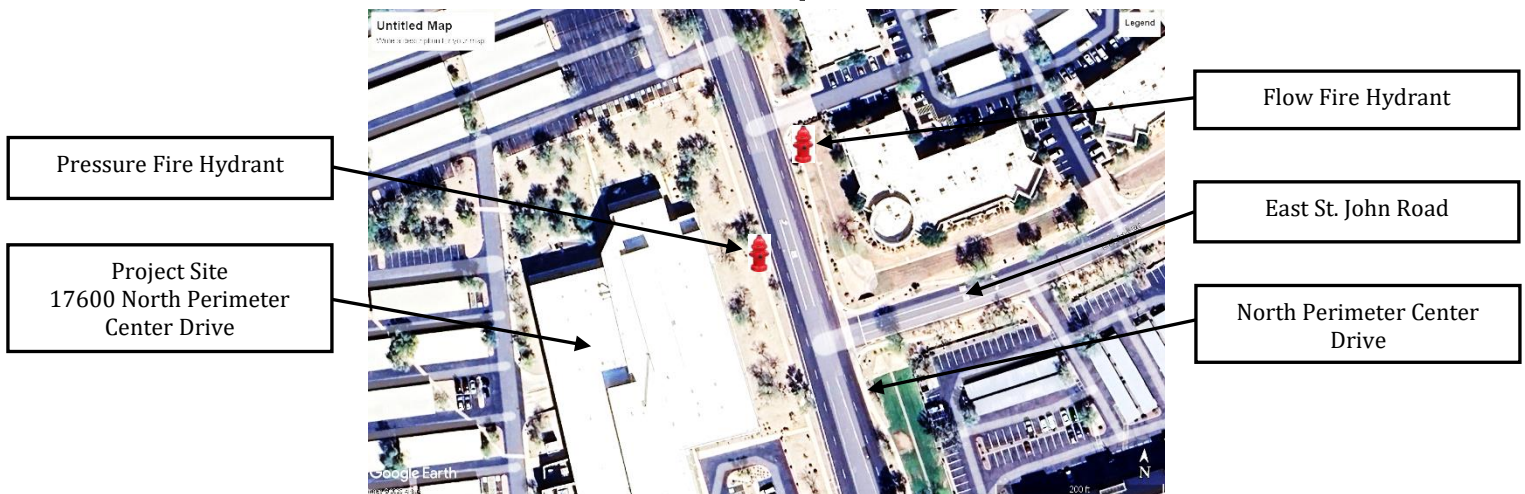
Flowing GPM: **2,946 GPM**

GPM @ 20 PSI: **6,503 GPM**

Scottsdale requires a maximum Static Pressure of 72 PSI for AFES Design.

### Flow Test Location

North ↑



**Appendix E**  
**Water Demand Exhibit**



**INSTRUCTIONS**

INPUT DEVELOPMENT NAME, CASE NUMBER, AND QUANTITY VALUES TO DETERMINE TOTAL AVERAGE DAILY WATER USE PER THE 2018 DESIGN STANDARDS AND POLICY MANUAL (DS7PM) CHAPTER 6 USING GALLONS PER DAY (GPD) VALUES FROM FIGURE 6-1.2

| TABLE 1: QUANTITY INPUT TABLE FOR THE DEVELOPMENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                   |                                                          |                       |                               |                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------|-----------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENTER DEVELOPMENT NAME & CITY REVIEW CASE NUMBER HERE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                   |                                                          |                       |                               |                                                                                                                                                                                                                                                              |
| WATER USE DEVELOPMENT TYPE/CATEGORY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AVERAGE UNIT WATER USE PER DS&PM CH. 6 (GPD/UNIT) | INPUT APPLICABLE QUANTITY FOR DEVELOPMENT IN THIS COLUMN | NUMERICAL UNIT        | TOTAL AVERAGE WATER USE (GPD) | NOTES                                                                                                                                                                                                                                                        |
| <b>Category: Residential/ Commerical Residential/ Hotel</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                   |                                                          |                       |                               |                                                                                                                                                                                                                                                              |
| < 2 DU/ac                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 485.6                                             |                                                          | DU                    | -                             | Community pool demands not included here. Refer to separate category.                                                                                                                                                                                        |
| 2 – 2.9 DU/ac                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 470.4                                             |                                                          | DU                    | -                             |                                                                                                                                                                                                                                                              |
| 3 – 7.9 DU/ac                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 248.2                                             |                                                          | DU                    | -                             |                                                                                                                                                                                                                                                              |
| 8 – 11.9 DU/ac                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 227.6                                             |                                                          | DU                    | -                             |                                                                                                                                                                                                                                                              |
| 12 – 22 DU/ac                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 227.6                                             |                                                          | DU                    | -                             |                                                                                                                                                                                                                                                              |
| High Density Condominium (condo)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 185.3                                             |                                                          | DU                    | -                             |                                                                                                                                                                                                                                                              |
| Resort Hotel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 446.3                                             |                                                          | ROOM                  | -                             | Includes site amenities such as 1 "standard" restaurant w/ associated dedicated kitchen, laundry service, landscaping, fountains, and 1 medium capacity pool. Large event venues/kitchens or multiple/large pools and multiple restaurants are not included. |
| <b>Category: Commerical/ Other</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                   |                                                          |                       |                               |                                                                                                                                                                                                                                                              |
| Restaurant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1.3                                               |                                                          | FT2                   | -                             |                                                                                                                                                                                                                                                              |
| Commercial/Retail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0.80                                              | 125,504                                                  | FT2                   | 100,403                       |                                                                                                                                                                                                                                                              |
| Commerical High Rise                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.60                                              |                                                          | FT2                   | -                             | per IBC highrise is at or over 75 feet to highest finished floor                                                                                                                                                                                             |
| Office                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.60                                              |                                                          | FT2                   | -                             |                                                                                                                                                                                                                                                              |
| Institutional                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1,340                                             |                                                          | ACRE                  | -                             |                                                                                                                                                                                                                                                              |
| Industrial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1,027                                             | 4                                                        | ACRE                  | 3,749                         |                                                                                                                                                                                                                                                              |
| Research and Development                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1,284                                             |                                                          | ACRE                  | -                             |                                                                                                                                                                                                                                                              |
| <b>Category: Special Use Areas</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                   |                                                          |                       |                               |                                                                                                                                                                                                                                                              |
| Natural Area Open Space                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -                                                 |                                                          | ACRE                  | -                             | Zero water demand                                                                                                                                                                                                                                            |
| Developed Open Space - Parks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1,786                                             |                                                          | ACRE                  | -                             |                                                                                                                                                                                                                                                              |
| Developed Open Space- Golf Course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4,285                                             |                                                          | ACRE                  | -                             |                                                                                                                                                                                                                                                              |
| <b>Category: Evaporation from Swimming Pools/Spas, Cooling, Turf Area Irrigation, Other Outdoor Consumptive Uses</b>                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                   |                                                          |                       |                               |                                                                                                                                                                                                                                                              |
| Extra large pool (60k to 100k gallons)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 274                                               |                                                          | EA                    | -                             | Annual mean ETo = 74.75 in as collected by AZ Met. Kc = 1.1. Average pool size of 400 sq. ft. loses 20,490 gallons per year, or 51.23 gallons per sq ft, not including backwashing or leaks, per AMWUA calculator.                                           |
| Large pool (above 30k to 60k gallons)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 154                                               |                                                          | EA                    | -                             |                                                                                                                                                                                                                                                              |
| Medium pool (15k to 30k gallons)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 75                                                |                                                          | EA                    | -                             |                                                                                                                                                                                                                                                              |
| Small pool or spa (under 15k gallons)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 51                                                | 1                                                        | EA                    | 51                            |                                                                                                                                                                                                                                                              |
| Total Bermuda Turf Area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.10                                              |                                                          | FT2                   | -                             | 1 sq ft of non-overseeded turf at 60% efficiency with increased Kc is 35 gallons per sq ft per year, per AMWUA calculator.                                                                                                                                   |
| Total Overseeded Turf Area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.02                                              |                                                          | FT2                   | -                             | 1 sq ft of overseeded turf at 60% efficiency with increased Kc is 9 gallons per sq ft per year, per AMWUA calculator.                                                                                                                                        |
| Evaporative Cooling/ Cooling Towers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | -                                                 |                                                          | TOTAL COOLING TONNAGE | -                             | Based on 1.50 cycles of concentration and average annual daily utilization of 68%. Water use is linear with respect to total cooling capacity tonnage. Based on US Dept of Energy Efficiency and Renewable Energy data.                                      |
| <b>Category: Filter Backwash Flows &amp; Make-up Water from Pools &amp; Spas (rapid sand filters)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                   |                                                          |                       |                               |                                                                                                                                                                                                                                                              |
| Extra large pool (60k to 100k gallons)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 229                                               | -                                                        | EA                    | -                             | Based on once per 7 day backwash @ 50,100, and 150gpm, respectively for each size pool category for 8 minute duration. Quantity values used from pool input values above.                                                                                    |
| Large pool (above 30k to 60k gallons)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 171                                               | -                                                        | EA                    | -                             |                                                                                                                                                                                                                                                              |
| Medium pool (15k to 30k gallons)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 114                                               | -                                                        | EA                    | -                             |                                                                                                                                                                                                                                                              |
| Small pool or spa (under 15k gallons)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 57                                                | 1                                                        | EA                    | 57                            |                                                                                                                                                                                                                                                              |
| <b>A. TOTAL AVERAGE DAILY WATER USE FOR THIS DEVELOPMENT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                   |                                                          |                       | <b>104,260</b>                | <b>GPD</b>                                                                                                                                                                                                                                                   |
| <b>NOTES:</b><br>GPD=GALLONS PER DAY, DU=DWELLING UNITS, FT2=SQUARE FEET, AC=ACRE, EA=EACH UNIT, ETo=EVAPOTRANSPIRATION, Kc=CROP COEFFICIENT, AZMET=ARIZONA METEOROLOGICAL NETWORK, AMWUA=ARIZONA MUNICIPAL WATER USERS ASSOCIATION<br>NONE OF THE VALUES OR CALCULATIONS HEREIN ARE INTENDED TO BE USED FOR INFRASTRUCTURE DESIGN, PEAK FLOW DETERMINATION, OR SYSTEM CAPACITY ANALYSIS. FOR THESE PURPOSES REFER TO CH.6 & 7 OF THE CITY'S DESIGN STANDARDS AND POLICY MANUAL FOR THE RESPECTIVE DESIGN VALUES AND PEAKING FACTORS. |                                                   |                                                          |                       |                               |                                                                                                                                                                                                                                                              |
| TABLE INPUT VALUES LAST UPDATED:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                   |                                                          |                       | 9/26/2025                     |                                                                                                                                                                                                                                                              |

**INSTRUCTIONS**

IDENTIFY WATER CONSERVATION MEASURES ABOVE THOSE REQUIRED BY CITY CODE THAT THE DEVELOPMENT(S) PROPOSE TO IMPLEMENT. ENTER AN "X" FOR EACH PROPOSED MEASURE. THIS TABLE IS THE INITIATION OF THIS PROCESS. ALL MEASURE REQUIRES APPROVAL OF THE WATER CONSERVATION OFFICE. CHOOSE OPTIONS THAT FIT THE DEVELOPEMENT. ITEM 7 AND 8 ARE REQUIREMENTS. ADD AN ATTACHMENT ELABORATING ON THE CONSERVATION ITEMS CHOOSEN.

| TABLE 2: APPROVED SUPPLEMENTAL WATER CONSERVATION MEASURES                                                                                                                                                                                                          |                                                              |                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FILL IN DEVELOPMENT NAME AND CITY CASE NUMBER ON TAB 1                                                                                                                                                                                                              |                                                              |                                                                                                                                                                               |
| PROPOSED FOR THIS DEVELOPMENT (ENTER "X")                                                                                                                                                                                                                           | MEASURE                                                      | DESCRIPTION                                                                                                                                                                   |
|                                                                                                                                                                                                                                                                     | 1. Submetering                                               | Multi-family and mixed-use developments SUBMETER UNITS for leak detection and for occupants ability to manage their own water use                                             |
|                                                                                                                                                                                                                                                                     | 2. No outdoor water features                                 | Decorative water features outdoors can be a source of water use that is not functional                                                                                        |
|                                                                                                                                                                                                                                                                     | 3. Indoor water features submetered                          | Water features have proven to be a source of leaks. Submetering that is capable of alerts to the building monitoring system greatly reduce water waste                        |
|                                                                                                                                                                                                                                                                     | 4. Limitations on artificial turf                            | Artificial turf is a large source of heat especially during summer months.                                                                                                    |
|                                                                                                                                                                                                                                                                     | 5. Landscaped Rainwater harvesting                           | Earthworks, such as berms and basins, are encouraged to promote passive rainwater harvesting for planned plants and trees                                                     |
|                                                                                                                                                                                                                                                                     | 6. Pools and splashpads submeters with monitoring technology | Pools and splashpad can be a source of leaks. Submetering that is capable of alerts to the building monitoring system greatly reduce water waste. Timers on Splash pads       |
| REQUIRED FOR ALL LWU DEVELOPMENT (ENTER "X")                                                                                                                                                                                                                        | MEASURE                                                      | DESCRIPTION                                                                                                                                                                   |
|                                                                                                                                                                                                                                                                     | 7. Cooling tower controllers with monitoring technology      | Arizona high evapotranspiration rates, cooling towers use significantly more water here than in other states. Monitory systems can optimize this water use.                   |
|                                                                                                                                                                                                                                                                     | 8. Limitation on functional turf grass                       | Functional grass turf are areas used for congregation of large number of people and should be limited to up to 10% of the landscapable area.                                  |
| OTHER (ENTER "X")                                                                                                                                                                                                                                                   | MEASURE                                                      | DESCRIPTION                                                                                                                                                                   |
|                                                                                                                                                                                                                                                                     | 9. Additional measures that are specific to the development  | Should there be a conversation measure the developer believes reduces water usage but is specific to the development. Discuss with the Water Conservation Office for approval |
| <b>NOTES:</b><br>Greywater systems and large areas of artificial turf are not recommended by water conservation.<br>This list represents water conservation measures that the Water Conservation Office has approved and has shown to provide proven water savings. |                                                              |                                                                                                                                                                               |
| TABLE INPUT VALUES LAST UPDATED:                                                                                                                                                                                                                                    |                                                              | 9/26/2025                                                                                                                                                                     |

## Water Demand Exhibit Summary

FILL IN DEVELOPMENT NAME AND CITY CASE NUMBER ON TAB 1

### 1. Total Estimated Water Use per Day on a Sustainable Basis (gallons per day, gpd)

104,260

gpd

### 2. Net Water (NW) / Consumptive Use (gallons per day, gpd)

26,820

gpd

### 3. Proposed Water Conservation Measures Above Those Required By City Code

|  |                                                              |              |
|--|--------------------------------------------------------------|--------------|
|  | 1. Submetering                                               | NOT PROPOSED |
|  | 2. No outdoor water features                                 | NOT PROPOSED |
|  | 3. Indoor water features submetered                          | NOT PROPOSED |
|  | 4. Limitations on artificial turf                            | NOT PROPOSED |
|  | 5. Landscaped Rainwater harvesting                           | NOT PROPOSED |
|  | 6. Pools and splashpads submeters with monitoring technology | NOT PROPOSED |
|  | 7. Cooling tower controllers with monitoring technology      | NOT PROPOSED |
|  | 8. Limitation on functional turf grass                       | NOT PROPOSED |
|  | 9. Additional measures that are specific to the development  | NOT PROPOSED |

### 4. Annual Economic Value of the Development on a per Gallon of Use Basis (Applies to Commercial or Mixed Use, To be Completed by City)

1. Major City Revenues

\$/1,000 gallons

2. Total Annual Output Impact

\$/1,000 gallons

TABLE INPUT VALUES LAST UPDATED:

9/26/2025

## TABLE 4: WATER USE SUMMARY

FILL IN DEVELOPMENT NAME AND CITY CASE NUMBER ON TAB 1

### WATER USE SUMMARY FOR THE DEVELOPMENT

| USE CATEGORY                              | AMOUNT         | UNITS      | % OF TOTAL USE | CALCULATION NOTES      |
|-------------------------------------------|----------------|------------|----------------|------------------------|
| <b>A. TOTAL DAILY AVERAGE WATER USE</b>   | <b>104,260</b> | <b>GPD</b> | <b>100.0%</b>  | A=B+C, C=D+E,<br>F=B+D |
| B. OUTDOOR CONSUMPTIVE USE                | 13,164         | GPD        | 12.6%          |                        |
| C. TOTAL INDOOR USE                       | 91,096         | GPD        | 87.4%          |                        |
| D. INDOOR CONSUMPTIVE USE                 | 13,656         | GPD        | 13.1%          |                        |
| E. WASTEWATER TO SEWER                    | 77,441         | GPD        | 74.3%          |                        |
| <b>F. TOTAL CONSUMPTIVE USE (NET USE)</b> | <b>26,820</b>  | <b>GPD</b> | <b>25.7%</b>   |                        |

**NOTES:**

GPD=GALLONS PER DAY

ALL VALUES ARE FOR AVERAGE WATER USE ANALYSIS ONLY. THIS CALCULATION IS NOT INTENDED TO BE USED FOR INFRASTRUCTURE DESIGN, PEAK FLOW DETERMINATION, OR SYSTEM CAPACITY ANALYSIS. FOR THESE PURPOSES REFER TO CH.6 & 7 OF THE CITY'S DESIGN STANDARDS AND POLICY MANUAL FOR THE RESPECTIVE DESIGN VALUES, PEAKING FACTORS, AND DESIGN REQUIREMENTS.

TOTAL AVERAGE WATER USE (GALLONS PER DAY, GPD)

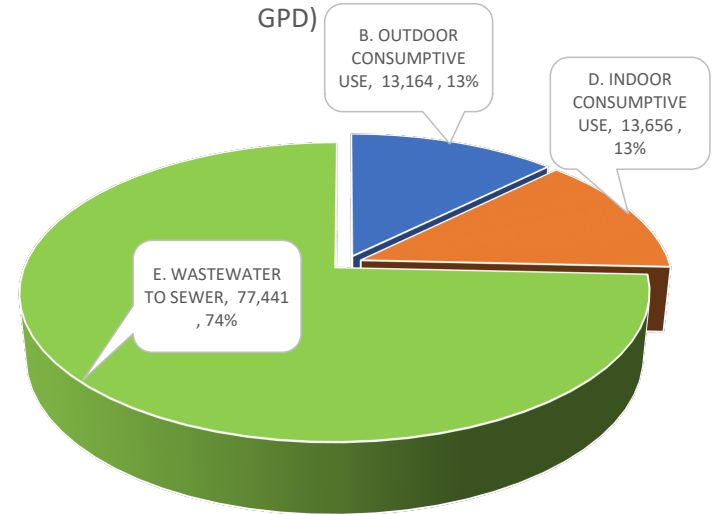


TABLE INPUT VALUES LAST UPDATED:

9/26/2025

TABLE 5: DETAILED WATER USE BREAKDOWN FOR THE DEVELOPMENT

FILL IN DEVELOPMENT NAME AND CITY CASE NUMBER ON TAB 1

| TO RIGHT:WATER USE ALLOCATION-<br>-->                                                                         | A. TOTAL<br>AVERAGE WATER<br>USE (GPD) | B. AVERAGE OUTDOOR <u>CONSUMPTIVE</u> WATER USE <sup>(1)</sup> |                                     |                                                | C. AVERAGE INDOOR <u>TOTAL</u> WATER USE <sup>(1)</sup> |                                  |                                         | D. AVERAGE INDOOR <u>CONSUMPTIVE</u> WATER USE <sup>(2)</sup> |                                           |                                                      | E. AVERAGE WASTEWATER FLOWS TO SEWER <sup>(3)</sup> |                          |                                |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------|-------------------------------------|------------------------------------------------|---------------------------------------------------------|----------------------------------|-----------------------------------------|---------------------------------------------------------------|-------------------------------------------|------------------------------------------------------|-----------------------------------------------------|--------------------------|--------------------------------|
| BELOW: WATER USE DEVELOPMENT<br>TYPE/CATEGORY                                                                 |                                        | UNIT OUTDOOR<br><u>CONSUMPTIVE</u><br>WATER USE<br>(GPD/UNIT)  | OUTDOOR<br>CONSUMPTIVE USE<br>(GPD) | OUTDOOR<br>CONSUMPTIVE USE<br>(% OF TOTAL USE) | UNIT <u>TOTAL</u><br>INDOOR WATER<br>USE (GPD/UNIT)     | INDOOR <u>TOTAL</u><br>USE (GPD) | INDOOR TOTAL<br>USE (% OF TOTAL<br>USE) | UNIT<br><u>CONSUMPTIVE</u><br>INDOOR WATER<br>USE (GPD/UNIT)  | INDOOR<br><u>CONSUMPTIVE</u> USE<br>(GPD) | INDOOR<br><u>CONSUMPTIVE</u> USE<br>(% OF TOTAL USE) | WASTEWATER<br>FLOW (GPD/UNIT)                       | WASTEWATER<br>FLOW (GPD) | WASTEWATER (%<br>OF TOTAL USE) |
| Category: Residential/ Commerical Residential/ Hotel                                                          |                                        |                                                                |                                     |                                                |                                                         |                                  |                                         |                                                               |                                           |                                                      |                                                     |                          |                                |
| < 2 DU/ac)                                                                                                    | -                                      | 276.7                                                          | -                                   | 0.0%                                           | 208.9                                                   | -                                | 0.0%                                    | 20.9                                                          | -                                         | 0.0%                                                 | 188                                                 | -                        | 0.0%                           |
| 2 – 2.9 DU/ac                                                                                                 | -                                      | 276.7                                                          | -                                   | 0.0%                                           | 193.7                                                   | -                                | 0.0%                                    | 19.4                                                          | -                                         | 0.0%                                                 | 174                                                 | -                        | 0.0%                           |
| 3 – 7.9 DU/ac                                                                                                 | -                                      | 72.3                                                           | -                                   | 0.0%                                           | 175.9                                                   | -                                | 0.0%                                    | 17.6                                                          | -                                         | 0.0%                                                 | 158                                                 | -                        | 0.0%                           |
| 8 – 11.9 DU/ac                                                                                                | -                                      | 72.3                                                           | -                                   | 0.0%                                           | 155.3                                                   | -                                | 0.0%                                    | 15.5                                                          | -                                         | 0.0%                                                 | 140                                                 | -                        | 0.0%                           |
| 12 – 22 DU/ac                                                                                                 | -                                      | 72.3                                                           | -                                   | 0.0%                                           | 155.3                                                   | -                                | 0.0%                                    | 15.5                                                          | -                                         | 0.0%                                                 | 140                                                 | -                        | 0.0%                           |
| High Density Condominium (condo)                                                                              | -                                      | 30.0                                                           | -                                   | 0.0%                                           | 155.3                                                   | -                                | 0.0%                                    | 15.5                                                          | -                                         | 0.0%                                                 | 140                                                 | -                        | 0.0%                           |
| Resort Hotel                                                                                                  | -                                      | 44.6                                                           | -                                   | 0.0%                                           | 401.7                                                   | -                                | 0.0%                                    | 32.1                                                          | -                                         | 0.0%                                                 | 370                                                 | -                        | 0.0%                           |
| Category: Commerical/ Other                                                                                   |                                        |                                                                |                                     |                                                |                                                         |                                  |                                         |                                                               |                                           |                                                      |                                                     |                          |                                |
| Restaurant                                                                                                    | -                                      | 0.10                                                           | -                                   | 0.0%                                           | 1.20                                                    | -                                | 0.0%                                    | 0.12                                                          | -                                         | 0.0%                                                 | 1.08                                                | -                        | 0.0%                           |
| Commercial/Retail                                                                                             | 100,403                                | 0.10                                                           | 12,550                              | 12.0%                                          | 0.70                                                    | 87,852.8                         | 84.3%                                   | 0.11                                                          | 13,177.9                                  | 12.6%                                                | 0.60                                                | 74,675                   | 71.6%                          |
| Commerical High Rise                                                                                          | -                                      | 0.10                                                           | -                                   | 0.0%                                           | 0.50                                                    | -                                | 0.0%                                    | 0.05                                                          | -                                         | 0.0%                                                 | 0.45                                                | -                        | 0.0%                           |
| Office                                                                                                        | -                                      | 0.10                                                           | -                                   | 0.0%                                           | 0.50                                                    | -                                | 0.0%                                    | 0.05                                                          | -                                         | 0.0%                                                 | 0.45                                                | -                        | 0.0%                           |
| Institutional                                                                                                 | -                                      | 670                                                            | -                                   | 0.0%                                           | 670.0                                                   | -                                | 0.0%                                    | 100.50                                                        | -                                         | 0.0%                                                 | 569.50                                              | -                        | 0.0%                           |
| Industrial                                                                                                    | 3,749                                  | 154                                                            | 562                                 | 0.5%                                           | 873.0                                                   | 3,186.5                          | 3.1%                                    | 130.95                                                        | 478.0                                     | 0.5%                                                 | 742.05                                              | 2,708                    | 2.6%                           |
| Research and Development                                                                                      | -                                      | 192                                                            | -                                   | 0.0%                                           | 1,092.0                                                 | -                                | 0.0%                                    | 163.80                                                        | -                                         | 0.0%                                                 | 928.20                                              | -                        | 0.0%                           |
| Category: Special Use Areas                                                                                   |                                        |                                                                |                                     |                                                |                                                         |                                  |                                         |                                                               |                                           |                                                      |                                                     |                          |                                |
| Natural Area Open Space                                                                                       | -                                      | -                                                              | -                                   | 0.0%                                           |                                                         |                                  |                                         |                                                               |                                           |                                                      | -                                                   | -                        | 0.0%                           |
| Developed Open Space - Parks                                                                                  | -                                      | 1,786                                                          | -                                   | 0.0%                                           |                                                         |                                  |                                         |                                                               |                                           |                                                      | -                                                   | -                        | 0.0%                           |
| Developed Open Space- Golf Course                                                                             | -                                      | 4,285                                                          | -                                   | 0.0%                                           |                                                         |                                  |                                         |                                                               |                                           |                                                      | -                                                   | -                        | 0.0%                           |
| Category: Evaporation from Swimming Pools/Spas, Cooling, Turf Area Irrigation, Other Outdoor Consumptive Uses |                                        |                                                                |                                     |                                                |                                                         |                                  |                                         |                                                               |                                           |                                                      |                                                     |                          |                                |
| Extra large pool (60k to 100k gallons)                                                                        | -                                      | 274                                                            | -                                   | 0.0%                                           |                                                         |                                  |                                         |                                                               |                                           |                                                      | -                                                   | -                        | 0.0%                           |
| Large pool (above 30k to 60k gallons)                                                                         | -                                      | 154                                                            | -                                   | 0.0%                                           |                                                         |                                  |                                         |                                                               |                                           |                                                      | -                                                   | -                        | 0.0%                           |
| Medium pool (15k to 30k gallons)                                                                              | -                                      | 75                                                             | -                                   | 0.0%                                           |                                                         |                                  |                                         |                                                               |                                           |                                                      | -                                                   | -                        | 0.0%                           |
| Small pool or spa (under 15k gallons)                                                                         | 51                                     | 51                                                             | 51                                  | 0.0%                                           |                                                         |                                  |                                         |                                                               |                                           |                                                      | -                                                   | -                        | 0.0%                           |
| Total Bermuda Turf Area                                                                                       | -                                      | 0.10                                                           | -                                   | 0.0%                                           |                                                         |                                  |                                         |                                                               |                                           |                                                      | -                                                   | -                        | 0.0%                           |
| Total Overseeded Turf Area                                                                                    | -                                      | 0.02                                                           | -                                   | 0.0%                                           |                                                         |                                  |                                         |                                                               |                                           |                                                      | -                                                   | -                        | 0.0%                           |
| Evaporative Cooling/ Cooling Towers                                                                           | -                                      | -                                                              | -                                   | 0.0%                                           |                                                         |                                  |                                         |                                                               |                                           |                                                      | -                                                   | -                        | 0.0%                           |
| Category: Filter Backwash Flows & Make-up Water from Pools & Spas (rapid sand filters)                        |                                        |                                                                |                                     |                                                |                                                         |                                  |                                         |                                                               |                                           |                                                      |                                                     |                          |                                |
| Extra large pool (60k to 100k gallons)                                                                        | -                                      |                                                                |                                     |                                                | 228.6                                                   | -                                | 0.0%                                    |                                                               |                                           |                                                      | 229                                                 | -                        | 0.0%                           |
| Large pool (above 30k to 60k gallons)                                                                         | -                                      |                                                                |                                     |                                                | 171.4                                                   | -                                | 0.0%                                    |                                                               |                                           |                                                      | 171                                                 | -                        | 0.0%                           |
| Medium pool (15k to 30k gallons)                                                                              | -                                      |                                                                |                                     |                                                | 114.3                                                   | -                                | 0.0%                                    |                                                               |                                           |                                                      | 114                                                 | -                        | 0.0%                           |
| Small pool or spa (under 15k gallons)                                                                         | 57                                     |                                                                |                                     |                                                | 57.1                                                    | 57.1                             | 0.1%                                    |                                                               |                                           |                                                      | 57                                                  | 57                       | 0.1%                           |
| TOTALS                                                                                                        | 104,260                                |                                                                | 13,164                              | 12.6%                                          |                                                         | 91,096                           | 87.4%                                   |                                                               | 13,656                                    | 13.1%                                                |                                                     | 77,441                   | 74.3%                          |

|                                                                      |  |            |                    |
|----------------------------------------------------------------------|--|------------|--------------------|
| F. TOTAL CONSUMPTIVE/NET WATER USE FOR<br>THIS DEVELOPMENT (B. + D.) |  | 26,820 GPD | 25.7% OF TOTAL USE |
|----------------------------------------------------------------------|--|------------|--------------------|

NOTES:  
(1) PER 2018 DS&PM CHAPTER 6, FIGURE 6-1.2  
(2) VARIES FROM 8% TO 15%, TYPICALLY 10%  
(3) WASTEWATER FLOWS TO SEWER ARE CALCULATED AS C. MINUS D.  
GPD=GALLONS PER DAY, DU=DWELLING UNIT, FT2=SQUARE FEET, AC=ACRE, EA=EACH UNIT  
NONE OF THE VALUES OR CALCULATIONS HEREIN ARE INTENDED TO BE USED FOR INFRASTRUCTURE DESIGN, PEAK FLOW DETERMINATION, OR SYSTEM CAPACITY ANALYSIS. FOR THESE PURPOSES REFER TO CH.6 & 7 OF THE CITY'S DESIGN STANDARDS AND POLICY MANUAL FOR THE RESPECTIVE DESIGN VALUES AND PEAKING FACTORS.

TABLE INPUT VALUES LAST UPDATED: 9/26/2025